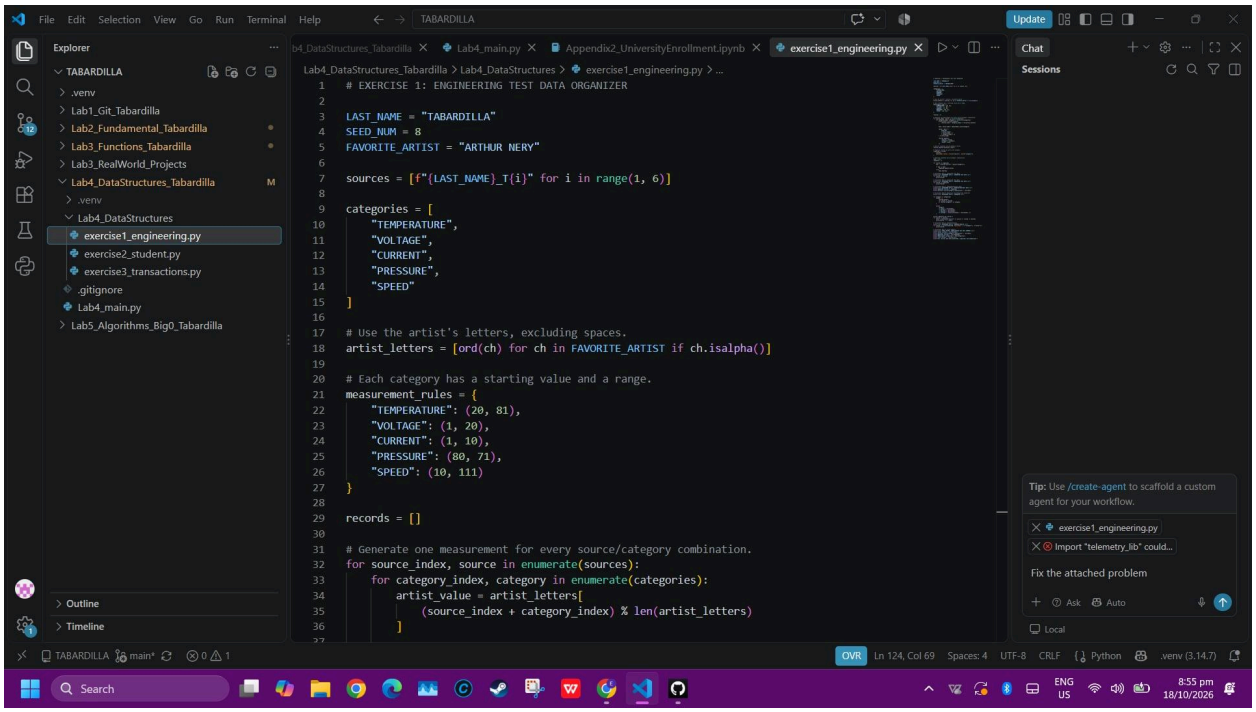
Erika S. Tabardilla
BSECE - 1C

Activity 4

EXERCISE 1: ENGINEERING TEST DATA ORGANIZER



```python
# EXERCISE 1: ENGINEERING TEST DATA ORGANIZER

LAST_NAME = "TABARDILLA"
SEED_NUM = 8
FAVORITE_ARTIST = "ARTHUR NERY"

sources = [f"{LAST_NAME}_T{i}" for i in range(1, 6)]

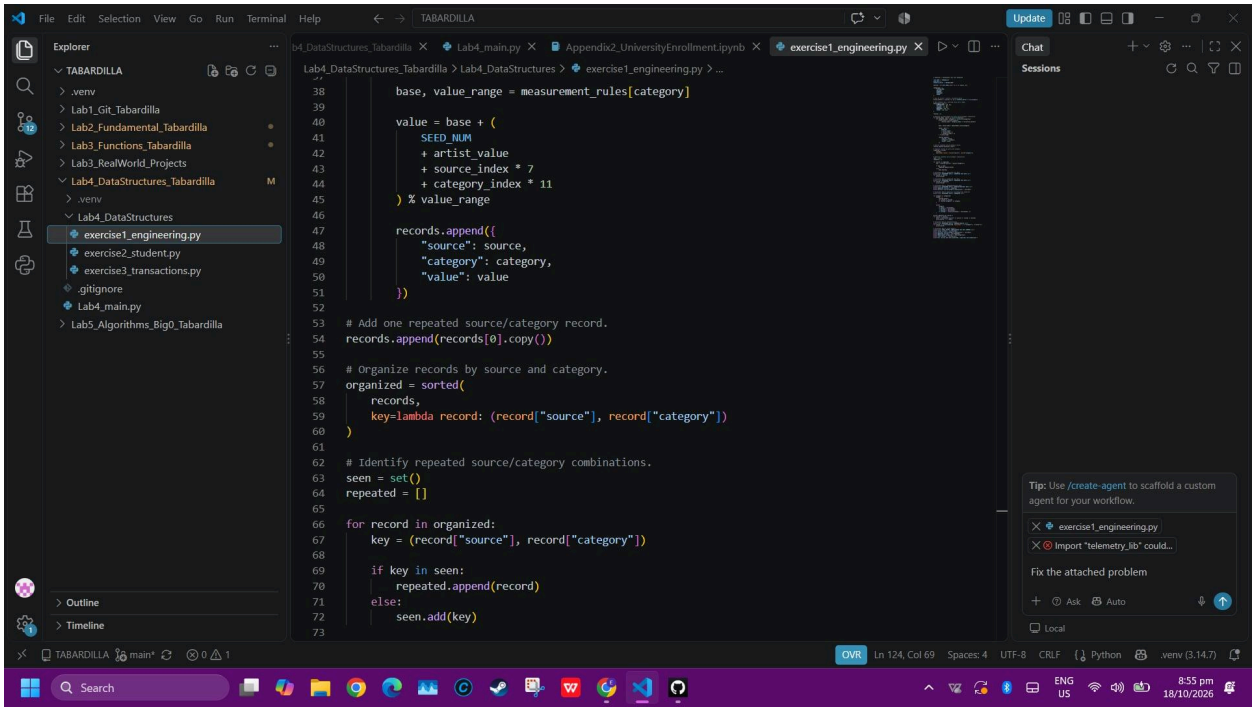
categories = [
    "TEMPERATURE",
    "VOLTAGE",
    "CURRENT",
    "PRESSURE",
    "SPEED"
]

# Use the artist's letters, excluding spaces.
artist_letters = [ord(ch) for ch in FAVORITE_ARTIST if ch.isalpha()]

# Each category has a starting value and a range.
measurement_rules = {
    "TEMPERATURE": (20, 81),
    "VOLTAGE": (1, 20),
    "CURRENT": (1, 10),
    "PRESSURE": (80, 71),
    "SPEED": (10, 111)
}

records = []

# Generate one measurement for every source/category combination.
for source_index, source in enumerate(sources):
    for category_index, category in enumerate(categories):
        artist_value = artist_letters[
            (source_index + category_index) % len(artist_letters)
        ]
```



```python
        base, value_range = measurement_rules[category]

        value = base + (
            SEED_NUM
            + artist_value
            + source_index * 7
            + category_index * 11
        ) % value_range

        records.append({
            "source": source,
            "category": category,
            "value": value
        })

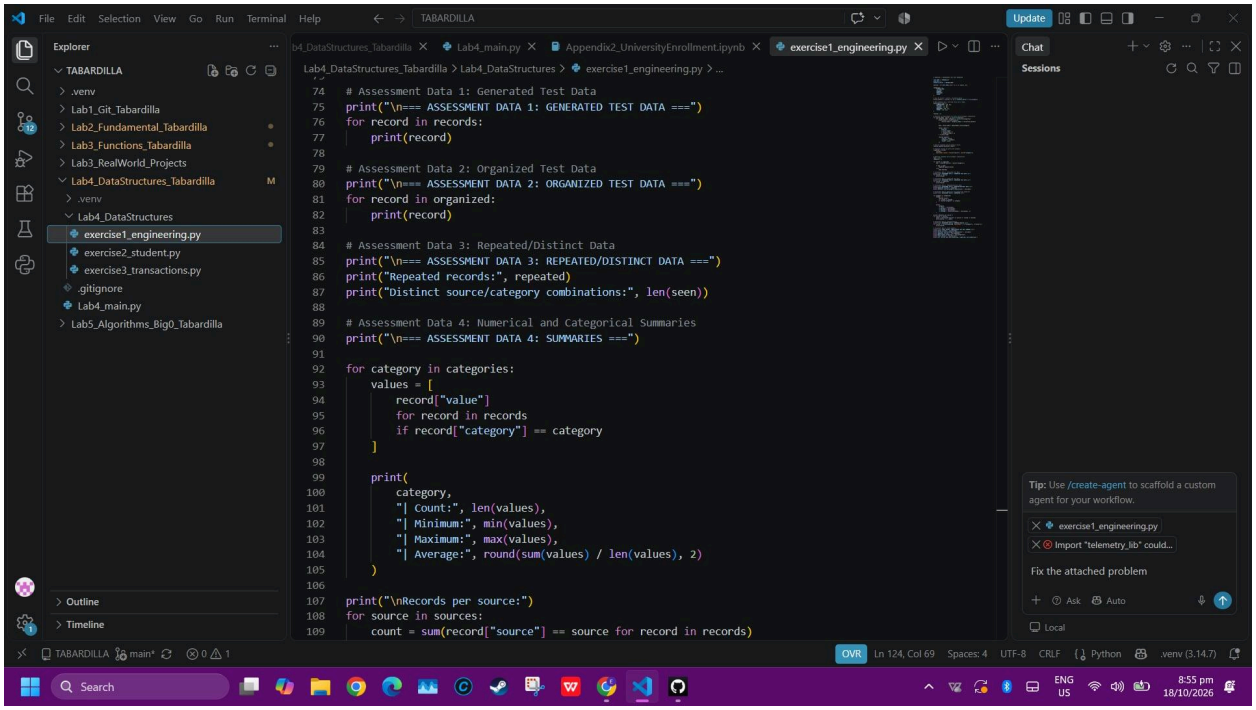
# Add one repeated source/category record.
records.append(records[0].copy())

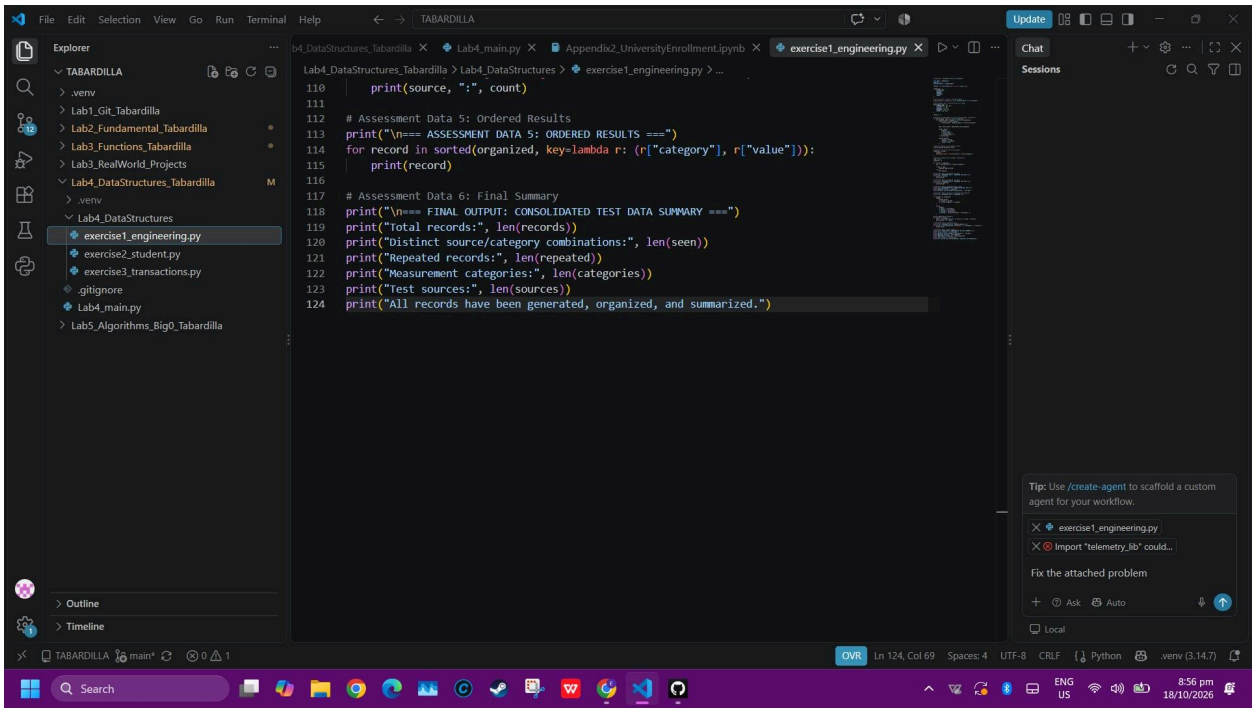
# Organize records by source and category.
organized = sorted(
    records,
    key=lambda record: (record["source"], record["category"])
)

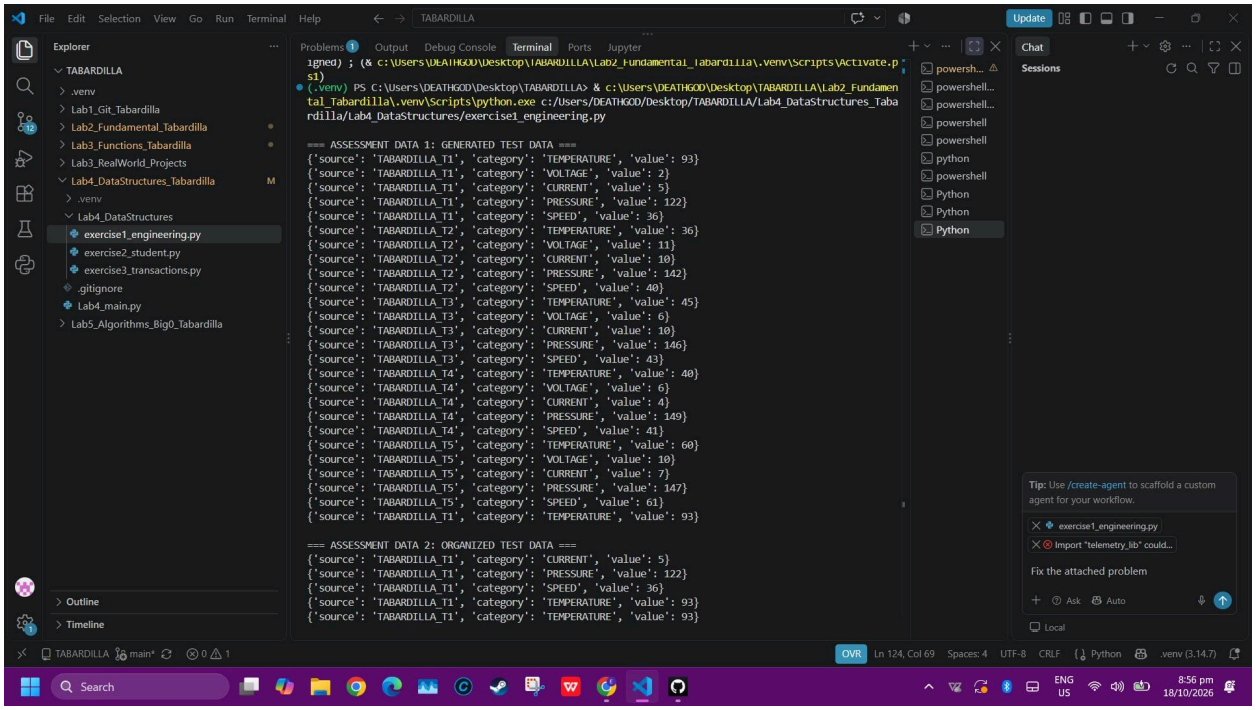
# Identify repeated source/category combinations.
seen = set()
repeated = []

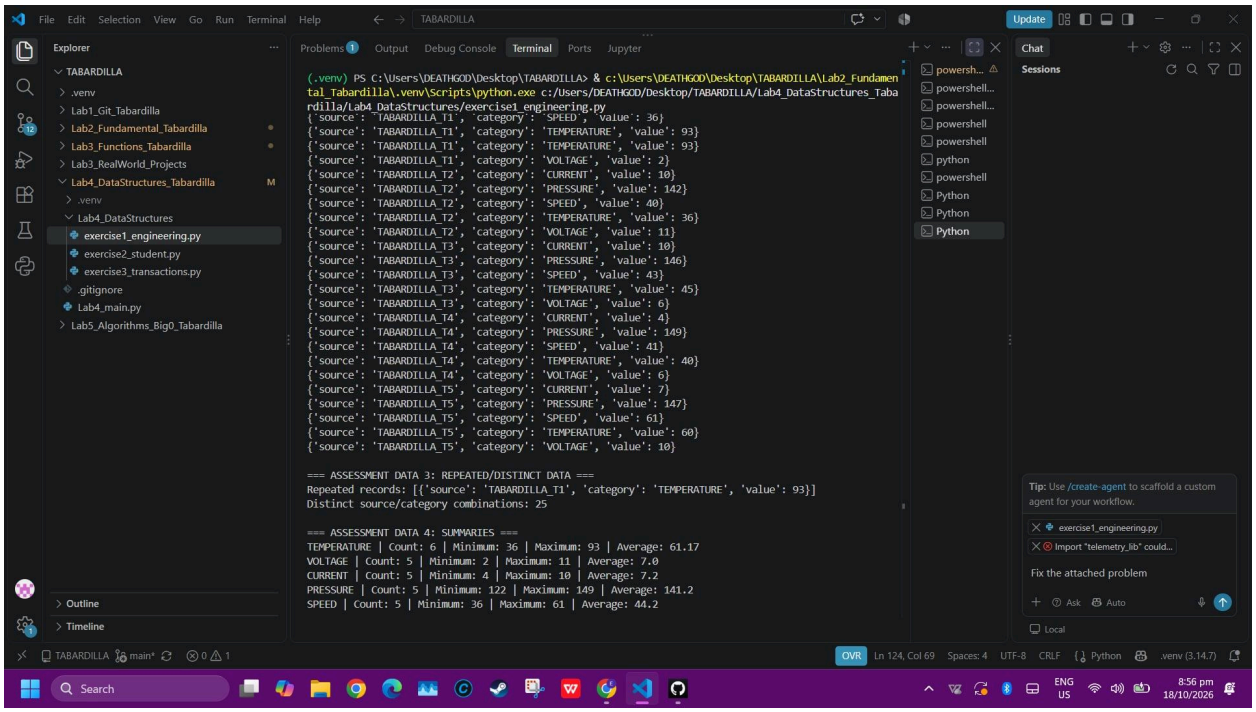
for record in organized:
    key = (record["source"], record["category"])

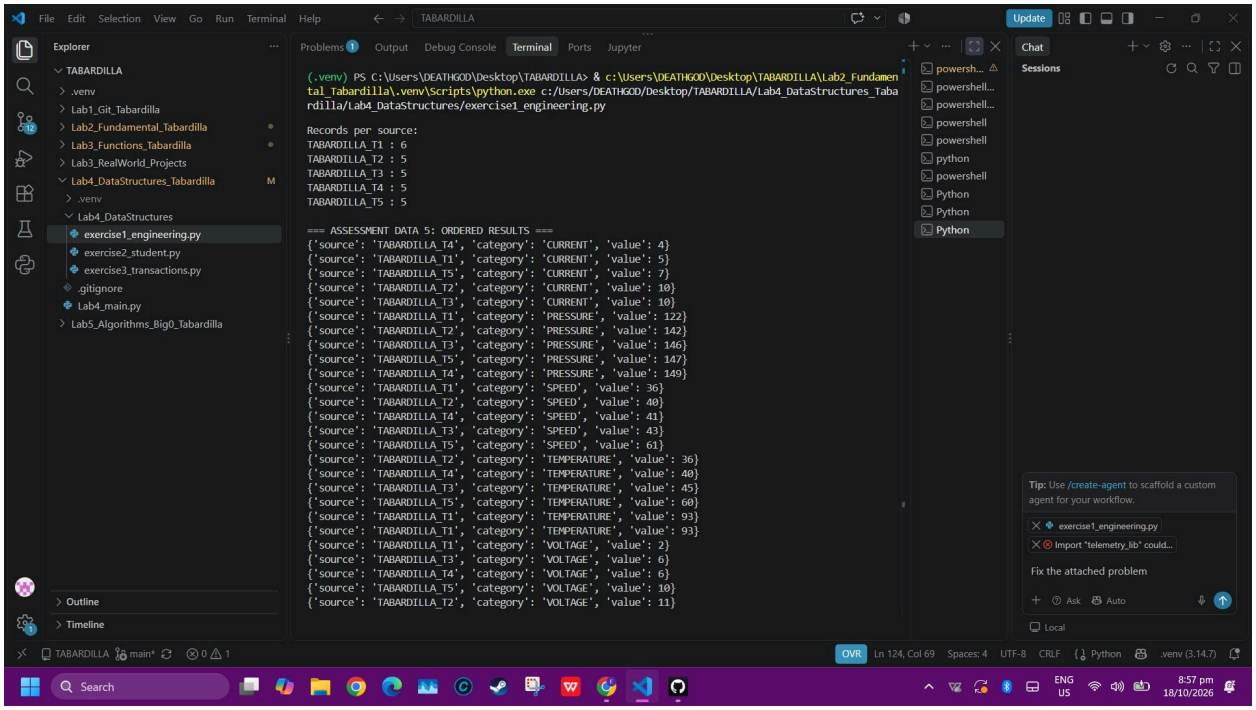
    if key in seen:
        repeated.append(record)
    else:
        seen.add(key)
```

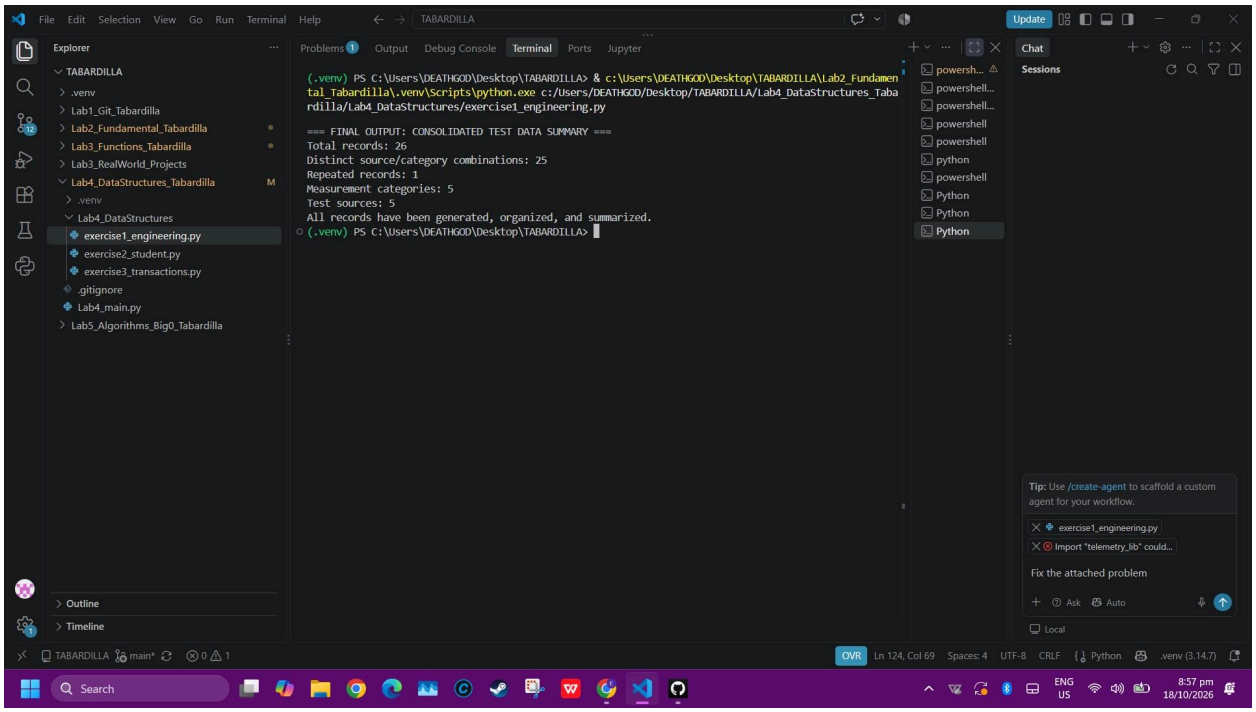

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Lab5_Algorithms_Big0_Tabardilla
Outline
Timeline
Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise1_engineering.py > ...
Assessment Data 1: Generated Test Data
print("\n=== ASSESSMENT DATA 1: GENERATED TEST DATA ===")
for record in records:
print(record)
Assessment Data 2: Organized Test Data
print("\n=== ASSESSMENT DATA 2: ORGANIZED TEST DATA ===")
for record in organized:
print(record)
Assessment Data 3: Repeated/Distinct Data
print("\n=== ASSESSMENT DATA 3: REPEATED/DISTINCT DATA ===")
print("Repeated records:", repeated)
print("Distinct source/category combinations:", len(seen))
Assessment Data 4: Numerical and Categorical Summaries
print("\n=== ASSESSMENT DATA 4: SUMMARIES ===")
for category in categories:
values = [
record["value"]
for record in records
if record["category"] == category
]
print(
category,
"| Count:", len(values),
"| Minimum:", min(values),
"| Maximum:", max(values),
"| Average:", round(sum(values) / len(values), 2)
)
print("\nRecords per source:")
for source in sources:
count = sum(record["source"] == source for record in records)
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise1_engineering.py
Import "telemetry_lib" could...
Fix the attached problem
Ask
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Search
ENG US
8:55 pm
18/10/2026


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Outline
Timeline
Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise1_engineering.py > ...
print(source, ":", count)
Assessment Data 5: Ordered Results
print("\n=== ASSESSMENT DATA 5: ORDERED RESULTS ===")
for record in sorted(organized, key=lambda r: (r["category"], r["value"])):
print(record)
Assessment Data 6: Final Summary
print("\n=== FINAL OUTPUT: CONSOLIDATED TEST DATA SUMMARY ===")
print("Total records:", len(records))
print("Distinct source/category combinations:", len(seen))
print("Repeated records:", len(repeated))
print("Measurement categories:", len(categories))
print("Test sources:", len(sources))
print("All records have been generated, organized, and summarized.")
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise1_engineering.py
Import "telemetry_lib" could...
Fix the attached problem
Ask
Auto
Local
TABARDILLA main*
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Spaces: 4
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18/10/2026

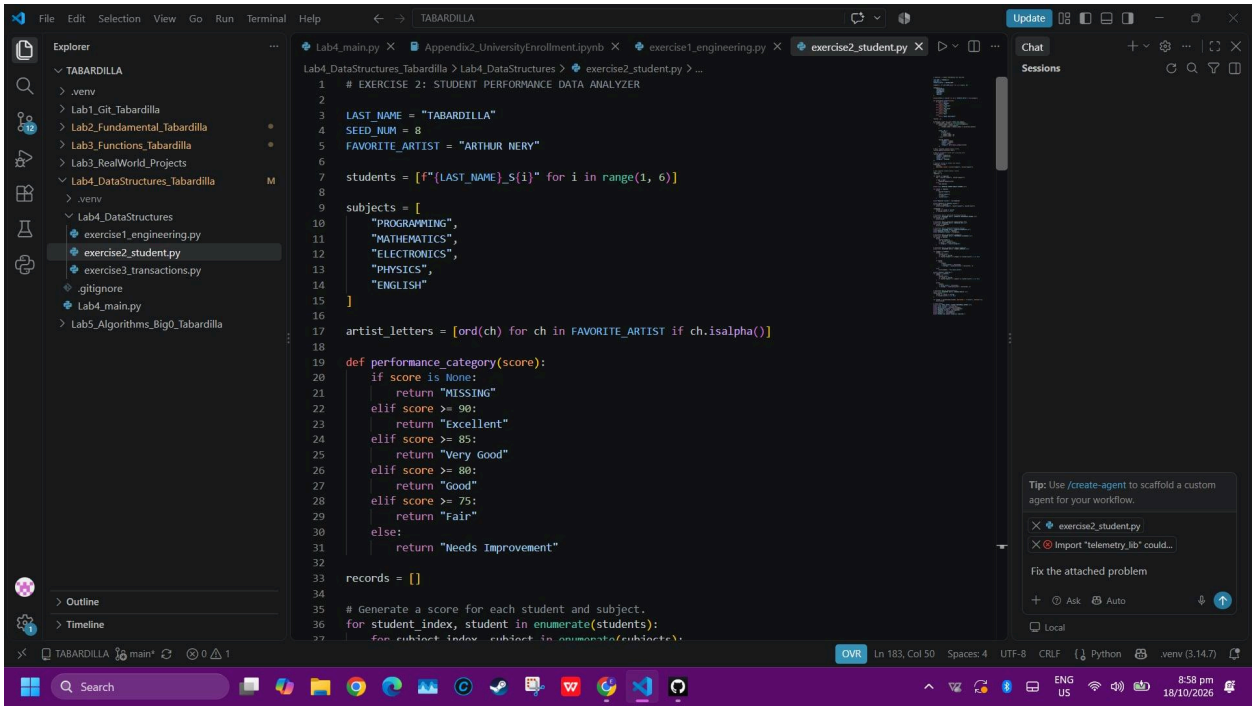

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Lab4_main.py
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Outline
Timeline
Problems 1 Output Debug Console Terminal Ports Jupyter
igned) ; (& c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA> & c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\python.exe c:/Users/DEATHGOD/Desktop/TABARDILLA/Lab4_DataStructures_Tabardilla/Lab4_DataStructures/exercise1_engineering.py
=== ASSESSMENT DATA 1: GENERATED TEST DATA ===
{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}
{'source': 'TABARDILLA_T1', 'category': 'VOLTAGE', 'value': 2}
{'source': 'TABARDILLA_T1', 'category': 'CURRENT', 'value': 5}
{'source': 'TABARDILLA_T1', 'category': 'PRESSURE', 'value': 122}
{'source': 'TABARDILLA_T1', 'category': 'SPEED', 'value': 36}
{'source': 'TABARDILLA_T2', 'category': 'TEMPERATURE', 'value': 36}
{'source': 'TABARDILLA_T2', 'category': 'VOLTAGE', 'value': 11}
{'source': 'TABARDILLA_T2', 'category': 'CURRENT', 'value': 10}
{'source': 'TABARDILLA_T2', 'category': 'PRESSURE', 'value': 142}
{'source': 'TABARDILLA_T2', 'category': 'SPEED', 'value': 40}
{'source': 'TABARDILLA_T3', 'category': 'TEMPERATURE', 'value': 45}
{'source': 'TABARDILLA_T3', 'category': 'VOLTAGE', 'value': 6}
{'source': 'TABARDILLA_T3', 'category': 'CURRENT', 'value': 10}
{'source': 'TABARDILLA_T3', 'category': 'PRESSURE', 'value': 146}
{'source': 'TABARDILLA_T3', 'category': 'SPEED', 'value': 43}
{'source': 'TABARDILLA_T4', 'category': 'TEMPERATURE', 'value': 40}
{'source': 'TABARDILLA_T4', 'category': 'VOLTAGE', 'value': 6}
{'source': 'TABARDILLA_T4', 'category': 'CURRENT', 'value': 4}
{'source': 'TABARDILLA_T4', 'category': 'PRESSURE', 'value': 149}
{'source': 'TABARDILLA_T4', 'category': 'SPEED', 'value': 41}
{'source': 'TABARDILLA_T5', 'category': 'TEMPERATURE', 'value': 60}
{'source': 'TABARDILLA_T5', 'category': 'VOLTAGE', 'value': 10}
{'source': 'TABARDILLA_T5', 'category': 'CURRENT', 'value': 7}
{'source': 'TABARDILLA_T5', 'category': 'PRESSURE', 'value': 147}
{'source': 'TABARDILLA_T5', 'category': 'SPEED', 'value': 61}
{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}
=== ASSESSMENT DATA 2: ORGANIZED TEST DATA ===
{'source': 'TABARDILLA_T1', 'category': 'CURRENT', 'value': 5}
{'source': 'TABARDILLA_T1', 'category': 'PRESSURE', 'value': 122}
{'source': 'TABARDILLA_T1', 'category': 'SPEED', 'value': 36}
{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}
{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}
powersh...
powershell...
powershell...
powershell
powershell
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powershell
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Python
Python
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise1_engineering.py
Import "telemetry_lib" could...
Fix the attached problem
Ask Auto
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OVR Ln 124, Col 69 Spaces: 4 UTF-8 CRLF Python .venv (3.14.7)
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ENG US
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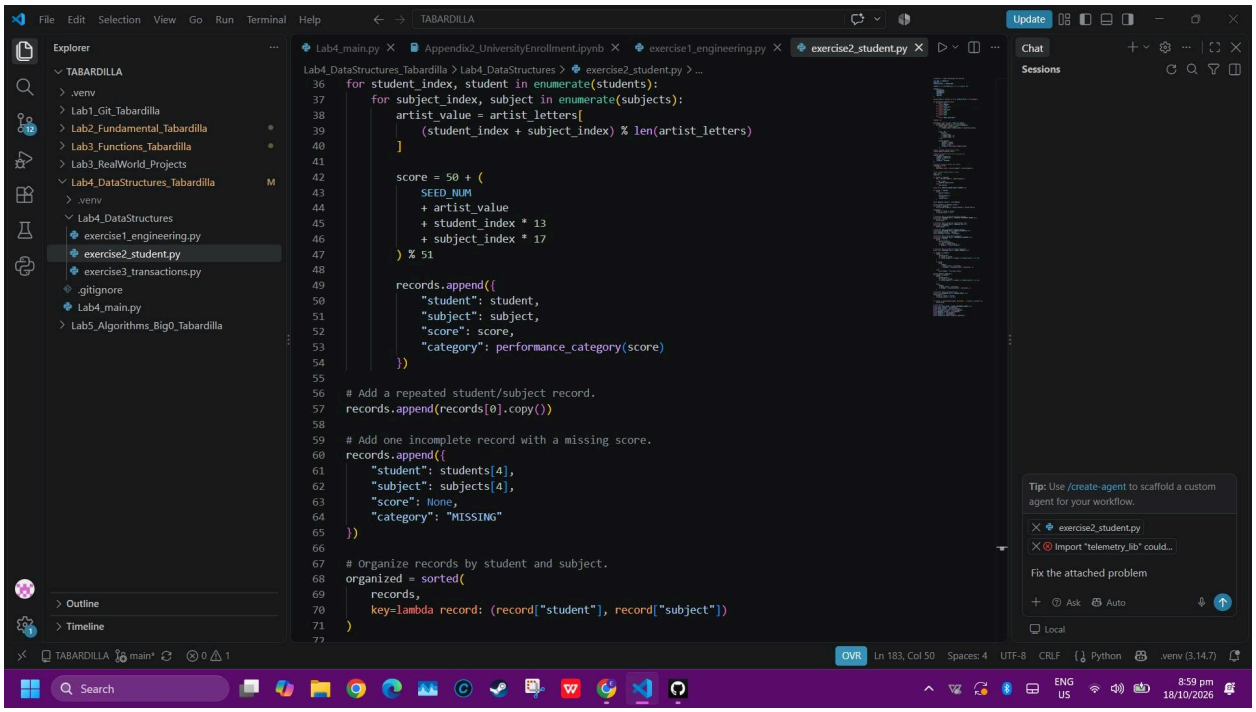

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Lab5_Algorithms_Big0_Tabardilla
Outline
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Problems 1 Output Debug Console Terminal Ports Jupyter
(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA> & c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\python.exe c:/Users/DEATHGOD/Desktop/TABARDILLA/Lab4_DataStructures_Tabardilla/Lab4_DataStructures/exercise1_engineering.py
{'source': 'TABARDILLA_T1', 'category': 'SPEED', 'value': 36}
{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}
{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}
{'source': 'TABARDILLA_T1', 'category': 'VOLTAGE', 'value': 2}
{'source': 'TABARDILLA_T2', 'category': 'CURRENT', 'value': 10}
{'source': 'TABARDILLA_T2', 'category': 'PRESSURE', 'value': 142}
{'source': 'TABARDILLA_T2', 'category': 'SPEED', 'value': 40}
{'source': 'TABARDILLA_T2', 'category': 'TEMPERATURE', 'value': 36}
{'source': 'TABARDILLA_T2', 'category': 'VOLTAGE', 'value': 11}
{'source': 'TABARDILLA_T3', 'category': 'CURRENT', 'value': 10}
{'source': 'TABARDILLA_T3', 'category': 'PRESSURE', 'value': 146}
{'source': 'TABARDILLA_T3', 'category': 'SPEED', 'value': 43}
{'source': 'TABARDILLA_T3', 'category': 'TEMPERATURE', 'value': 45}
{'source': 'TABARDILLA_T3', 'category': 'VOLTAGE', 'value': 6}
{'source': 'TABARDILLA_T4', 'category': 'CURRENT', 'value': 4}
{'source': 'TABARDILLA_T4', 'category': 'PRESSURE', 'value': 149}
{'source': 'TABARDILLA_T4', 'category': 'SPEED', 'value': 41}
{'source': 'TABARDILLA_T4', 'category': 'TEMPERATURE', 'value': 40}
{'source': 'TABARDILLA_T4', 'category': 'VOLTAGE', 'value': 6}
{'source': 'TABARDILLA_T5', 'category': 'CURRENT', 'value': 7}
{'source': 'TABARDILLA_T5', 'category': 'PRESSURE', 'value': 147}
{'source': 'TABARDILLA_T5', 'category': 'SPEED', 'value': 61}
{'source': 'TABARDILLA_T5', 'category': 'TEMPERATURE', 'value': 60}
{'source': 'TABARDILLA_T5', 'category': 'VOLTAGE', 'value': 10}
=== ASSESSMENT DATA 3: REPEATED/DISTINCT DATA ===
Repeated records: [{'source': 'TABARDILLA_T1', 'category': 'TEMPERATURE', 'value': 93}]
Distinct source/category combinations: 25
=== ASSESSMENT DATA 4: SUMMARIES ===
TEMPERATURE | Count: 6 | Minimum: 36 | Maximum: 93 | Average: 61.17
VOLTAGE | Count: 5 | Minimum: 2 | Maximum: 11 | Average: 7.0
CURRENT | Count: 5 | Minimum: 4 | Maximum: 10 | Average: 7.2
PRESSURE | Count: 5 | Minimum: 122 | Maximum: 149 | Average: 141.2
SPEED | Count: 5 | Minimum: 36 | Maximum: 61 | Average: 44.2
powersh...
powershell...
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Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise1_engineering.py
Import "telemetry_lib" could...
Fix the attached problem
Ask Auto
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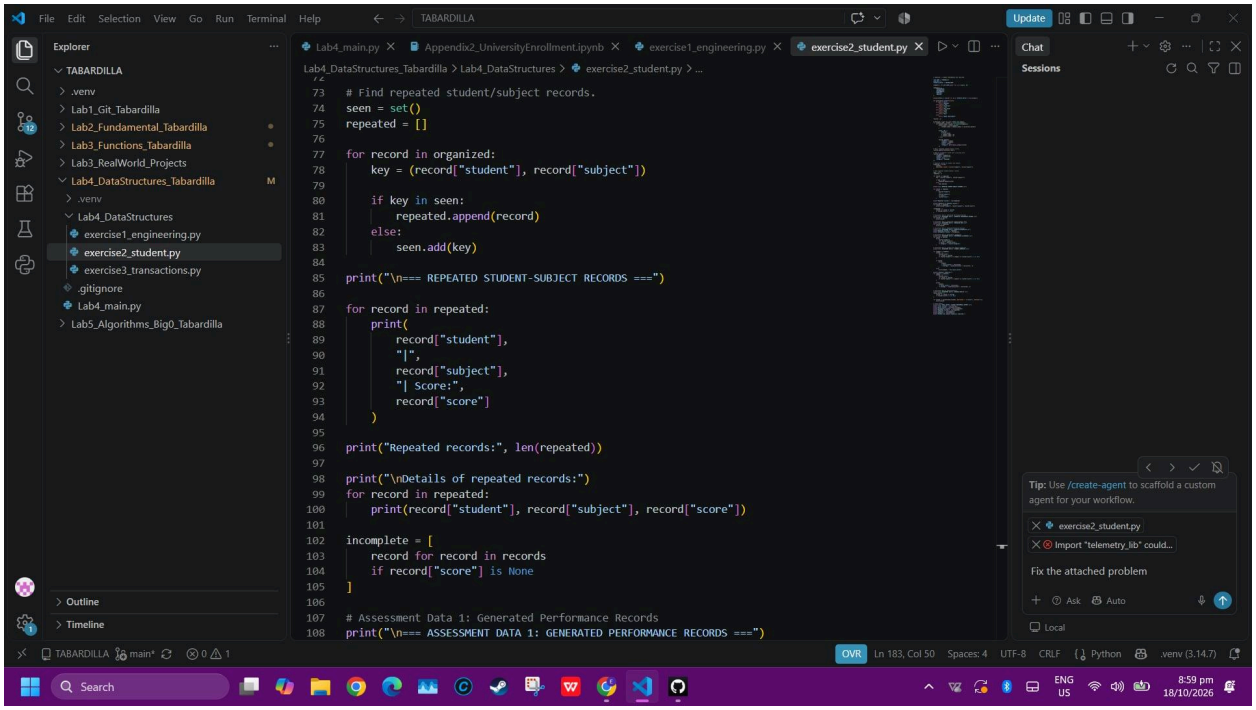


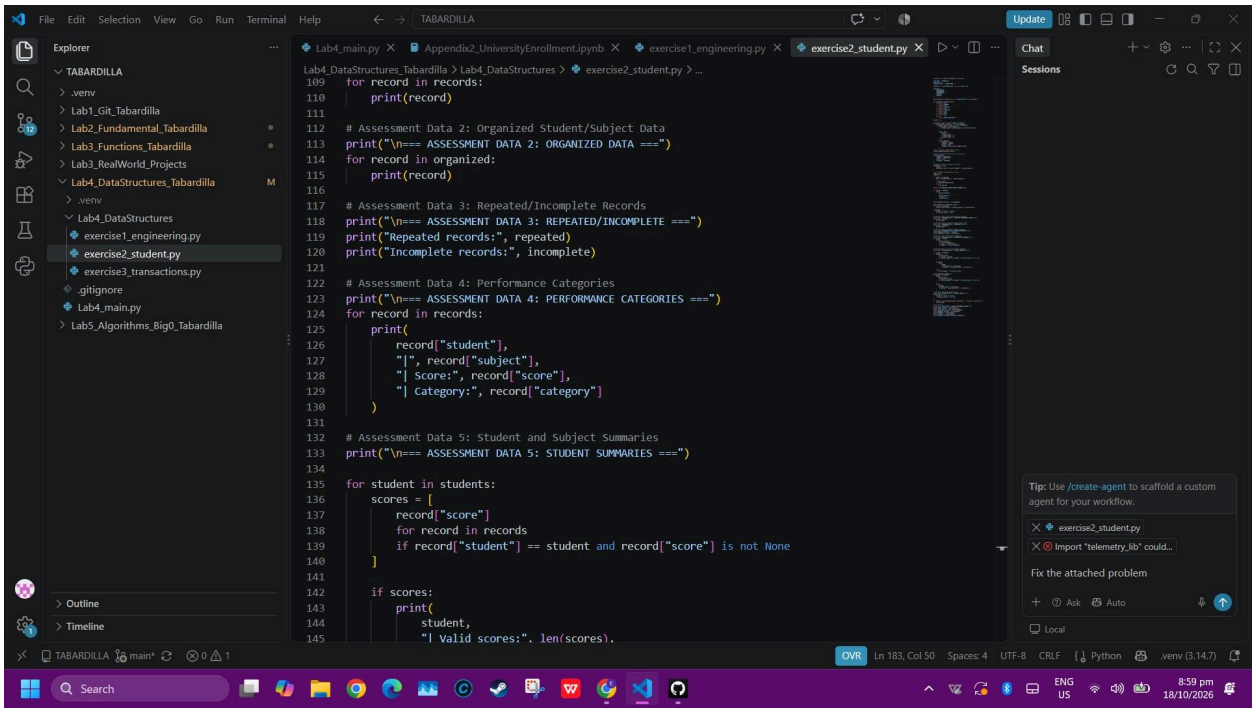


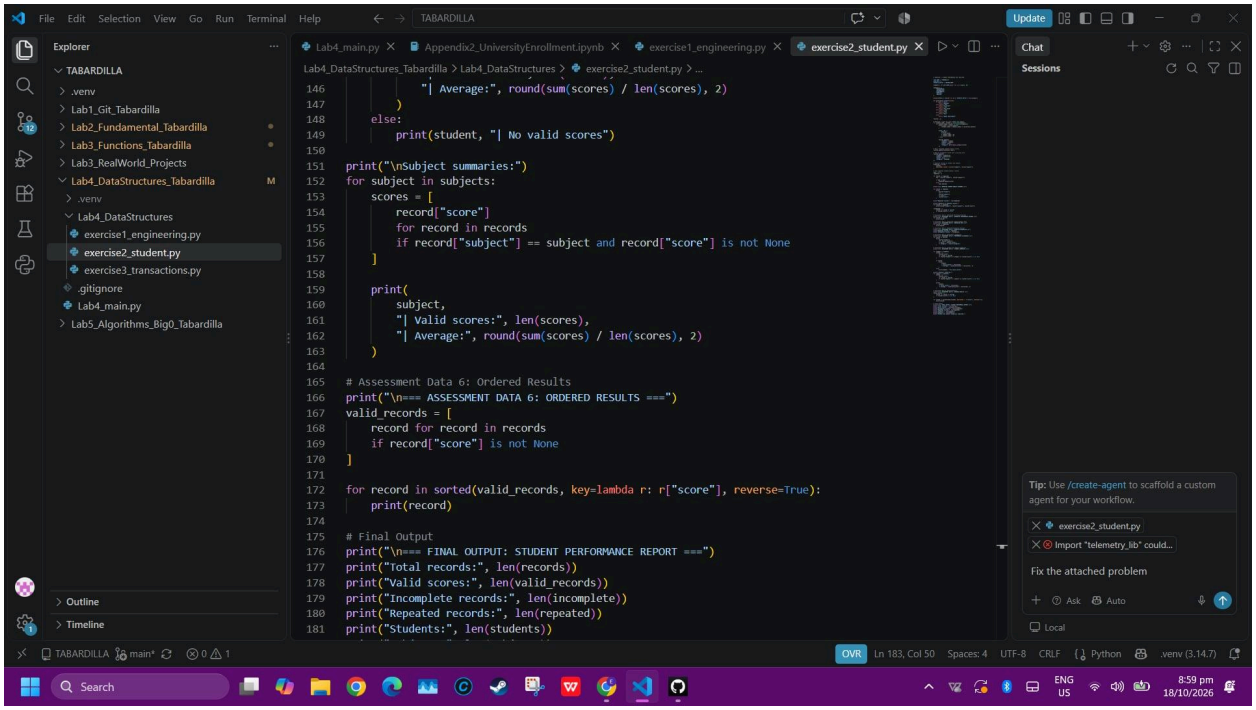
EXERCISE 2: STUDENT PERFORMANCE DATA ANALYZER

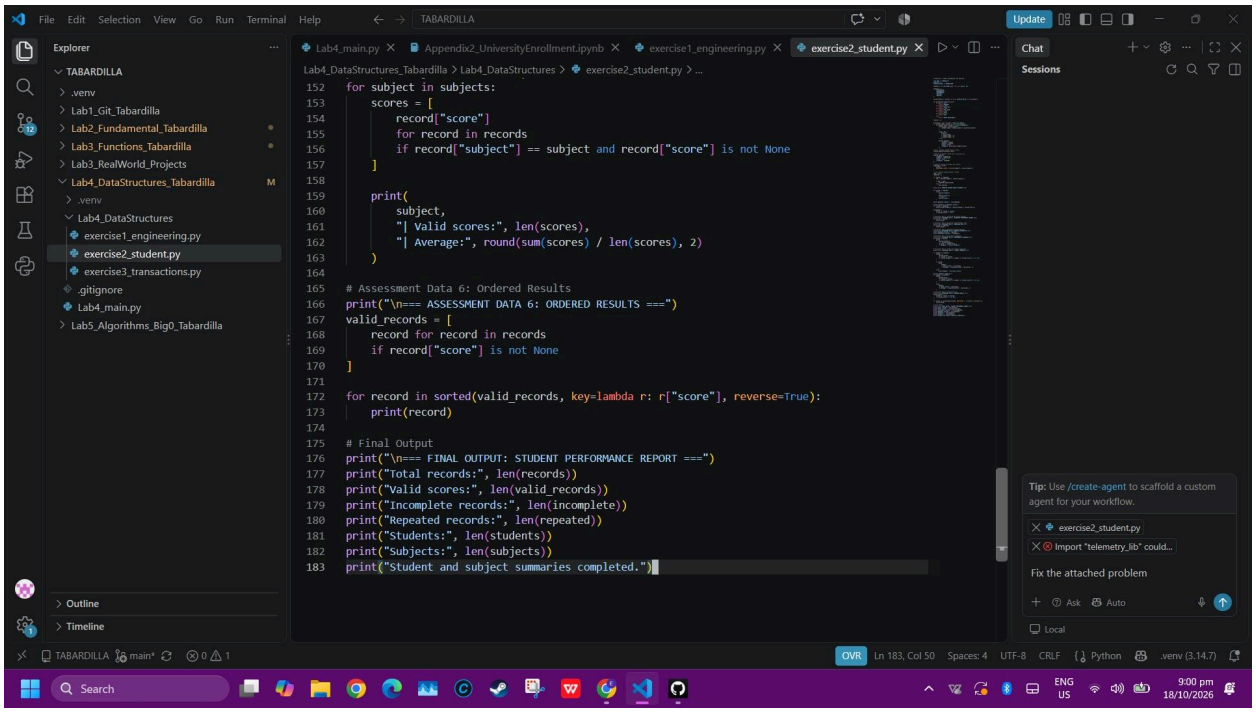

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Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise2_student.py > ...
EXERCISE 2: STUDENT PERFORMANCE DATA ANALYZER
LAST_NAME = "TABARDILLA"
SEED_NUM = 8
FAVORITE_ARTIST = "ARTHUR NERY"
students = [f"{LAST_NAME}_S{i}" for i in range(1, 6)]
subjects = [
"PROGRAMMING",
"MATHEMATICS",
"ELECTRONICS",
"PHYSICS",
"ENGLISH"
]
artist_letters = [ord(ch) for ch in FAVORITE_ARTIST if ch.isalpha()]
def performance_category(score):
if score is None:
return "MISSING"
elif score >= 90:
return "Excellent"
elif score >= 85:
return "Very Good"
elif score >= 80:
return "Good"
elif score >= 75:
return "Fair"
else:
return "Needs Improvement"
records = []
Generate a score for each student and subject.
for student_index, student in enumerate(students):
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise2_student.py
Import "telemetry_lib" could...
Fix the attached problem
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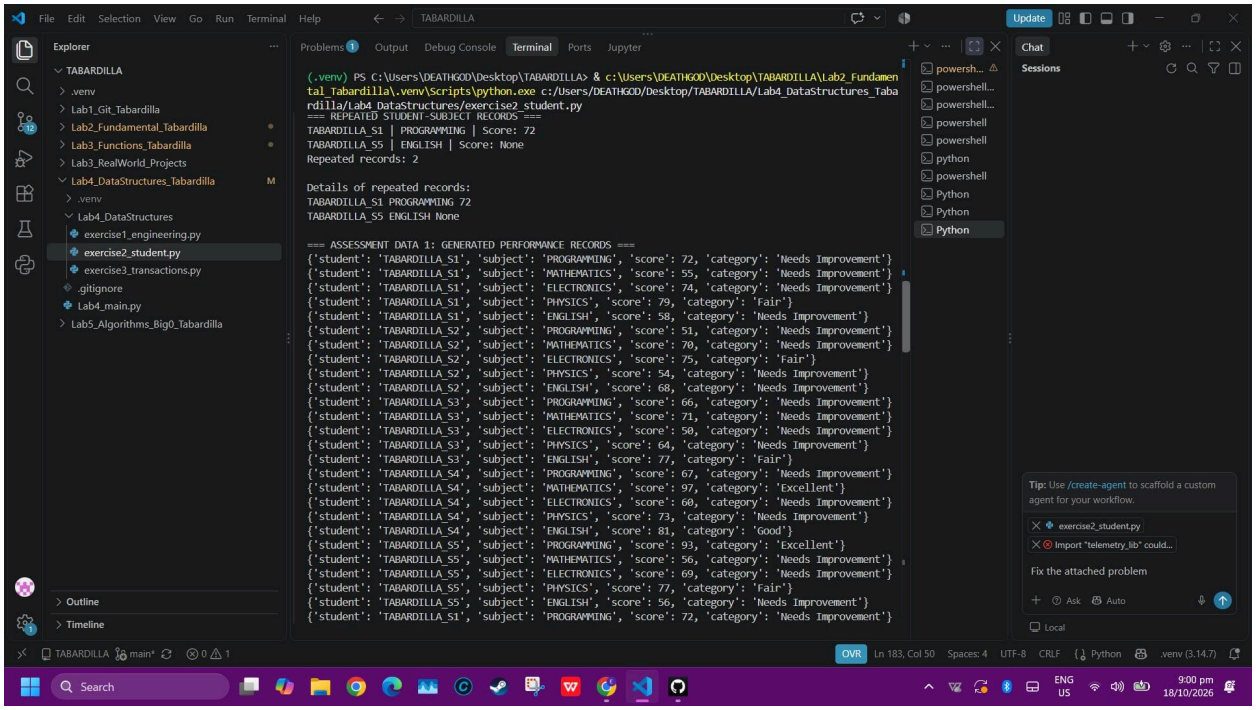

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exercise2_student.py
Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise2_student.py > ...
for student_index, student in enumerate(students):
for subject_index, subject in enumerate(subjects):
artist_value = artist_letters[
(student_index + subject_index) % len(artist_letters)
]
score = 50 + (
SEED_NUM
+ artist_value
+ student_index * 13
+ subject_index * 17
) % 51
records.append({
"student": student,
"subject": subject,
"score": score,
"category": performance_category(score)
})
Add a repeated student/subject record.
records.append(records[0].copy())
Add one incomplete record with a missing score.
records.append({
"student": students[4],
"subject": subjects[4],
"score": None,
"category": "MISSING"
})
Organize records by student and subject.
organized = sorted(
records,
key=lambda record: (record["student"], record["subject"])
)
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise2_student.py
Import "telemetry_lib" could...
Fix the attached problem
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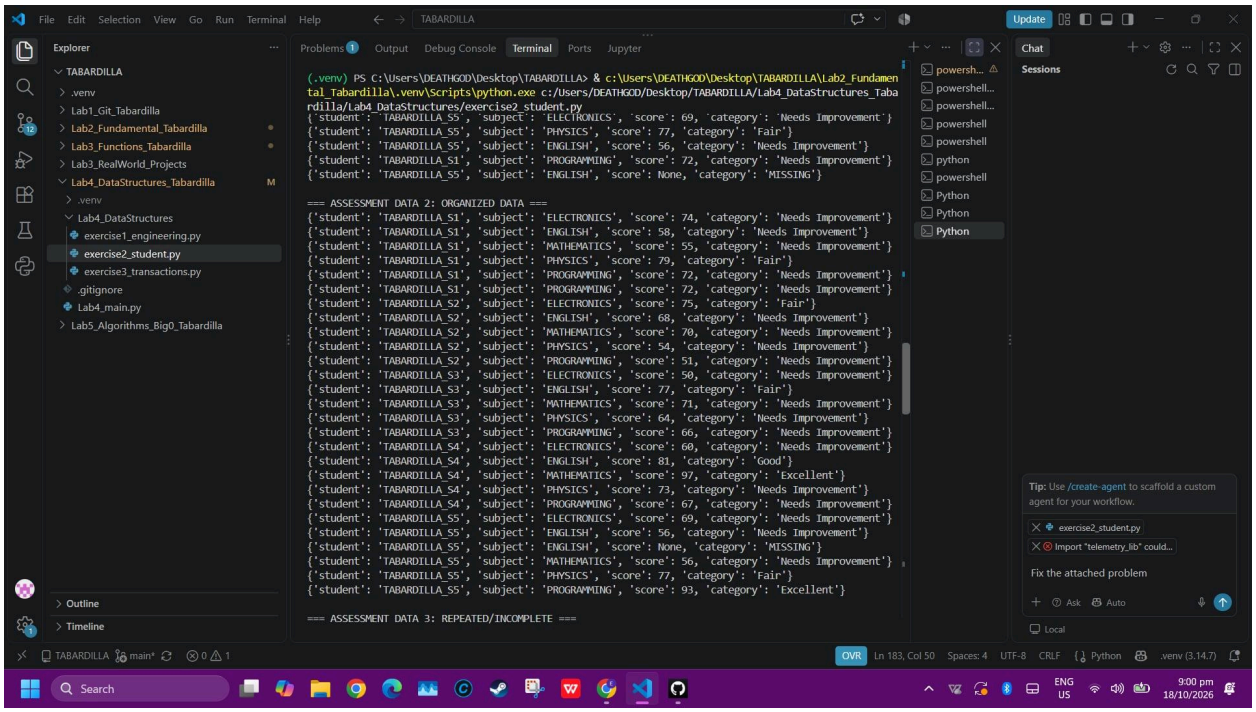

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Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise2_student.py > ...
Find repeated student/subject records.
seen = set()
repeated = []
for record in organized:
key = (record["student"], record["subject"])
if key in seen:
repeated.append(record)
else:
seen.add(key)
print("\n=== REPEATED STUDENT-SUBJECT RECORDS ===")
for record in repeated:
print(
record["student"],
"|",
record["subject"],
"| Score:",
record["score"]
)
print("Repeated records:", len(repeated))
print("\nDetails of repeated records:")
for record in repeated:
print(record["student"], record["subject"], record["score"])
incomplete = [
record for record in records
if record["score"] is None
]
Assessment Data 1: Generated Performance Records
print("\n=== ASSESSMENT DATA 1: GENERATED PERFORMANCE RECORDS ===")
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise2_student.py
Import "telemetry_lib" could...
Fix the attached problem
Ask
Auto
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TABARDILLA main*
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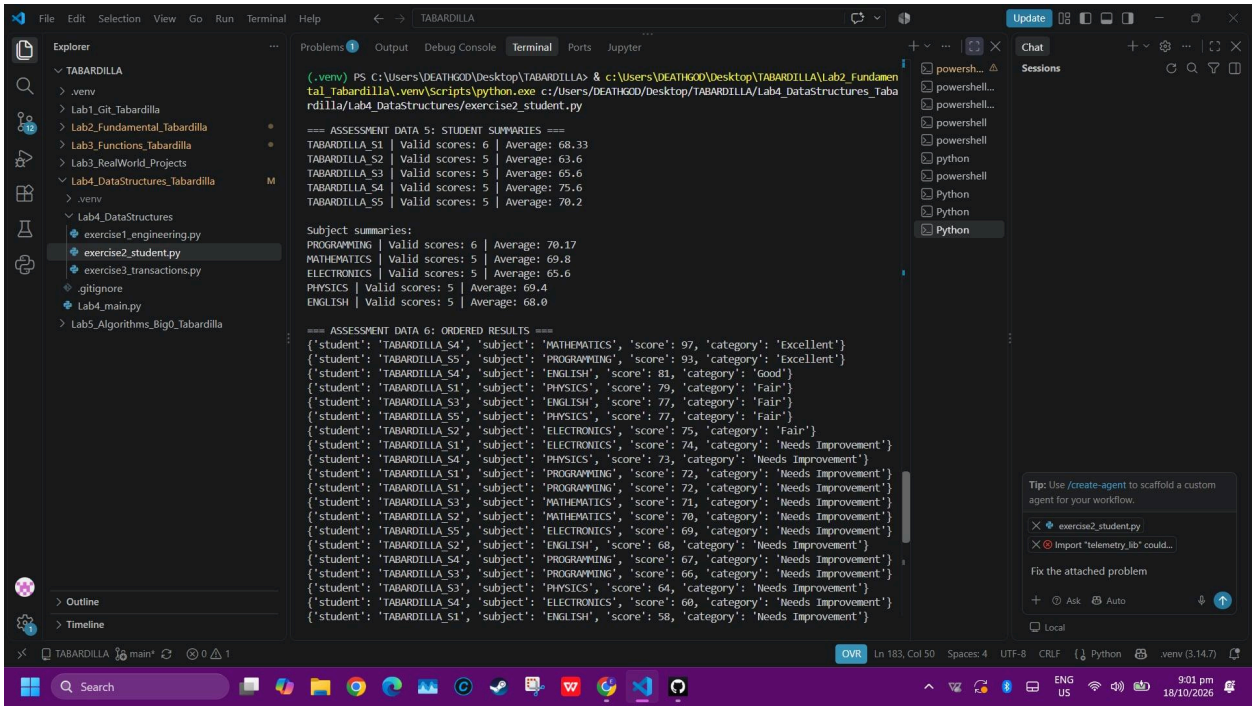

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Appendix2_UniversityEnrollment.ipynb
exercise1_engineering.py
exercise2_student.py
Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise2_student.py > ...
for record in records:
print(record)
Assessment Data 2: Organized Student/Subject Data
print("\n=== ASSESSMENT DATA 2: ORGANIZED DATA ===")
for record in organized:
print(record)
Assessment Data 3: Repeated/Incomplete Records
print("\n=== ASSESSMENT DATA 3: REPEATED/INCOMPLETE ===")
print("Repeated records:", repeated)
print("Incomplete records:", incomplete)
Assessment Data 4: Performance Categories
print("\n=== ASSESSMENT DATA 4: PERFORMANCE CATEGORIES ===")
for record in records:
print(
record["student"],
"|", record["subject"],
"| Score:", record["score"],
"| Category:", record["category"]
)
Assessment Data 5: Student and Subject Summaries
print("\n=== ASSESSMENT DATA 5: STUDENT SUMMARIES ===")
for student in students:
scores = [
record["score"]
for record in records
if record["student"] == student and record["score"] is not None
]
if scores:
print(
student,
"| Valid scores:", len(scores),
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise2_student.py
Import "telemetry_lib" could...
Fix the attached problem
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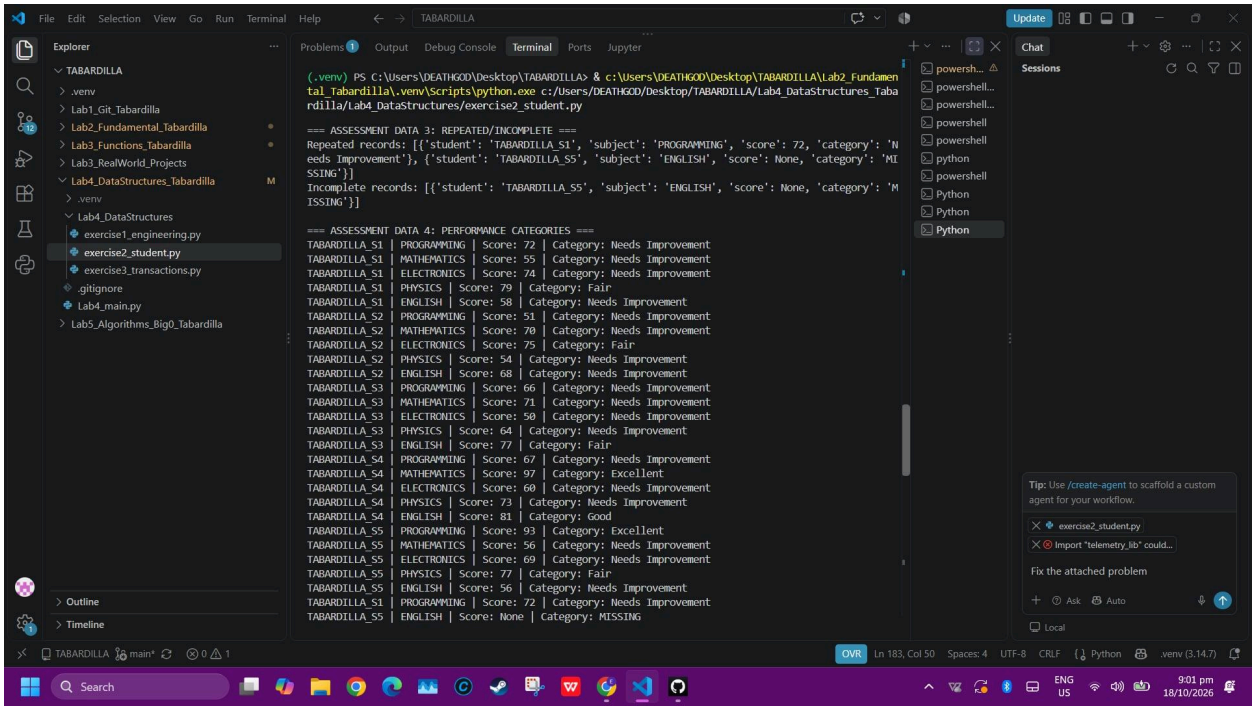

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Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise2_student.py > ...
"| Average:", round(sum(scores) / len(scores), 2)
)
else:
print(student, "| No valid scores")
print("\nSubject summaries:")
for subject in subjects:
scores = [
record["score"]
for record in records
if record["subject"] == subject and record["score"] is not None
]
print(
subject,
"| Valid scores:", len(scores),
"| Average:", round(sum(scores) / len(scores), 2)
)
Assessment Data 6: Ordered Results
print("\n=== ASSESSMENT DATA 6: ORDERED RESULTS ===")
valid_records = [
record for record in records
if record["score"] is not None
]
for record in sorted(valid_records, key=lambda r: r["score"], reverse=True):
print(record)
Final Output
print("\n=== FINAL OUTPUT: STUDENT PERFORMANCE REPORT ===")
print("Total records:", len(records))
print("Valid scores:", len(valid_records))
print("Incomplete records:", len(incomplete))
print("Repeated records:", len(repeated))
print("Students:", len(students))
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
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Import "telemetry_lib" could...
Fix the attached problem
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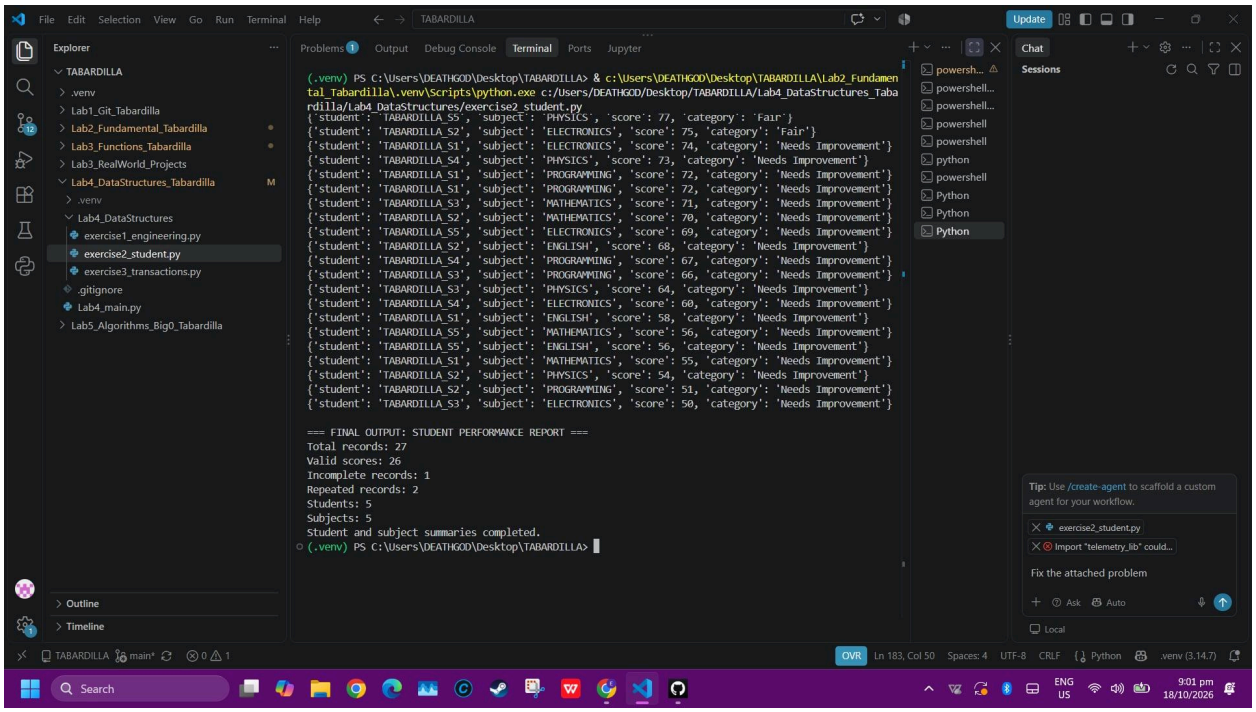

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for subject in subjects:
scores = [
record["score"]
for record in records
if record["subject"] == subject and record["score"] is not None
]
print(
subject,
"| Valid scores:", len(scores),
"| Average:", round(sum(scores) / len(scores), 2)
)
Assessment Data 6: Ordered Results
print("\n=== ASSESSMENT DATA 6: ORDERED RESULTS ===")
valid_records = [
record for record in records
if record["score"] is not None
]
for record in sorted(valid_records, key=lambda r: r["score"], reverse=True):
print(record)
Final Output
print("\n=== FINAL OUTPUT: STUDENT PERFORMANCE REPORT ===")
print("Total records:", len(records))
print("Valid scores:", len(valid_records))
print("Incomplete records:", len(incomplete))
print("Repeated records:", len(repeated))
print("Students:", len(students))
print("Subjects:", len(subjects))
print("Student and subject summaries completed.")
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise2_student.py
Import "telemetry_lib" could...
Fix the attached problem
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(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA> & c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\python.exe c:/Users/DEATHGOD/Desktop/TABARDILLA/Lab4_DataStructures_Tabardilla/Lab4_DataStructures/exercise2_student.py
=== REPEATED STUDENT-SUBJECT RECORDS ===
TABARDILLA_S1 | PROGRAMMING | Score: 72
TABARDILLA_S5 | ENGLISH | Score: None
Repeated records: 2
Details of repeated records:
TABARDILLA_S1 PROGRAMMING 72
TABARDILLA_S5 ENGLISH None
=== ASSESSMENT DATA 1: GENERATED PERFORMANCE RECORDS ===
{'student': 'TABARDILLA_S1', 'subject': 'PROGRAMMING', 'score': 72, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'MATHEMATICS', 'score': 55, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'ELECTRONICS', 'score': 74, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'PHYSICS', 'score': 79, 'category': 'Fair'}
{'student': 'TABARDILLA_S1', 'subject': 'ENGLISH', 'score': 58, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'PROGRAMMING', 'score': 51, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'MATHEMATICS', 'score': 70, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'ELECTRONICS', 'score': 75, 'category': 'Fair'}
{'student': 'TABARDILLA_S2', 'subject': 'PHYSICS', 'score': 54, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'ENGLISH', 'score': 68, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'MATHEMATICS', 'score': 71, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'ELECTRONICS', 'score': 50, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'PHYSICS', 'score': 64, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'ENGLISH', 'score': 77, 'category': 'Fair'}
{'student': 'TABARDILLA_S4', 'subject': 'PROGRAMMING', 'score': 67, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S4', 'subject': 'MATHEMATICS', 'score': 97, 'category': 'Excellent'}
{'student': 'TABARDILLA_S4', 'subject': 'ELECTRONICS', 'score': 60, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S4', 'subject': 'PHYSICS', 'score': 73, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S4', 'subject': 'ENGLISH', 'score': 81, 'category': 'Good'}
{'student': 'TABARDILLA_S5', 'subject': 'PROGRAMMING', 'score': 93, 'category': 'Excellent'}
{'student': 'TABARDILLA_S5', 'subject': 'MATHEMATICS', 'score': 56, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'ELECTRONICS', 'score': 69, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'PHYSICS', 'score': 77, 'category': 'Fair'}
{'student': 'TABARDILLA_S5', 'subject': 'ENGLISH', 'score': 56, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'PROGRAMMING', 'score': 72, 'category': 'Needs Improvement'}
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Fix the attached problem

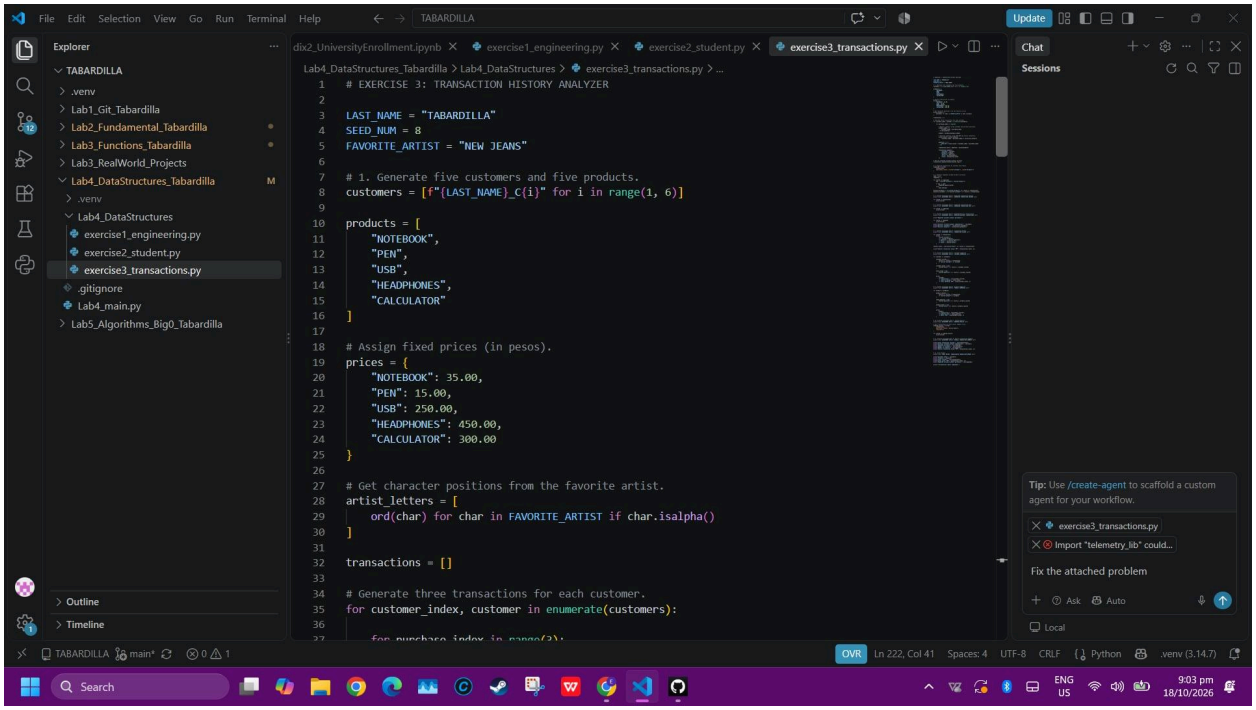

(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA> & c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\python.exe c:/Users/DEATHGOD/Desktop/TABARDILLA/Lab4_DataStructures_Tabardilla/Lab4_DataStructures/exercise2_student.py
{'student': 'TABARDILLA_S5', 'subject': 'ELECTRONICS', 'score': 69, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'PHYSICS', 'score': 77, 'category': 'Fair'}
{'student': 'TABARDILLA_S5', 'subject': 'ENGLISH', 'score': 56, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'PROGRAMMING', 'score': 72, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'ENGLISH', 'score': None, 'category': 'MISSING'}
=== ASSESSMENT DATA 2: ORGANIZED DATA ===
{'student': 'TABARDILLA_S1', 'subject': 'ELECTRONICS', 'score': 74, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'ENGLISH', 'score': 58, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'MATHEMATICS', 'score': 55, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'PHYSICS', 'score': 79, 'category': 'Fair'}
{'student': 'TABARDILLA_S1', 'subject': 'PROGRAMMING', 'score': 72, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'PROGRAMMING', 'score': 72, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'ELECTRONICS', 'score': 75, 'category': 'Fair'}
{'student': 'TABARDILLA_S2', 'subject': 'ENGLISH', 'score': 68, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'MATHEMATICS', 'score': 70, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'PHYSICS', 'score': 54, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S2', 'subject': 'PROGRAMMING', 'score': 51, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'ELECTRONICS', 'score': 50, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'ENGLISH', 'score': 77, 'category': 'Fair'}
{'student': 'TABARDILLA_S3', 'subject': 'MATHEMATICS', 'score': 71, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'PHYSICS', 'score': 64, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S3', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S4', 'subject': 'ELECTRONICS', 'score': 60, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S4', 'subject': 'ENGLISH', 'score': 81, 'category': 'Good'}
{'student': 'TABARDILLA_S4', 'subject': 'MATHEMATICS', 'score': 97, 'category': 'Excellent'}
{'student': 'TABARDILLA_S4', 'subject': 'PHYSICS', 'score': 73, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S4', 'subject': 'PROGRAMMING', 'score': 67, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'ELECTRONICS', 'score': 69, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'ENGLISH', 'score': 56, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'ENGLISH', 'score': None, 'category': 'MISSING'}
{'student': 'TABARDILLA_S5', 'subject': 'MATHEMATICS', 'score': 56, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S5', 'subject': 'PHYSICS', 'score': 77, 'category': 'Fair'}
{'student': 'TABARDILLA_S5', 'subject': 'PROGRAMMING', 'score': 93, 'category': 'Excellent'}
=== ASSESSMENT DATA 3: REPEATED/INCOMPLETE ===
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Fix the attached problem

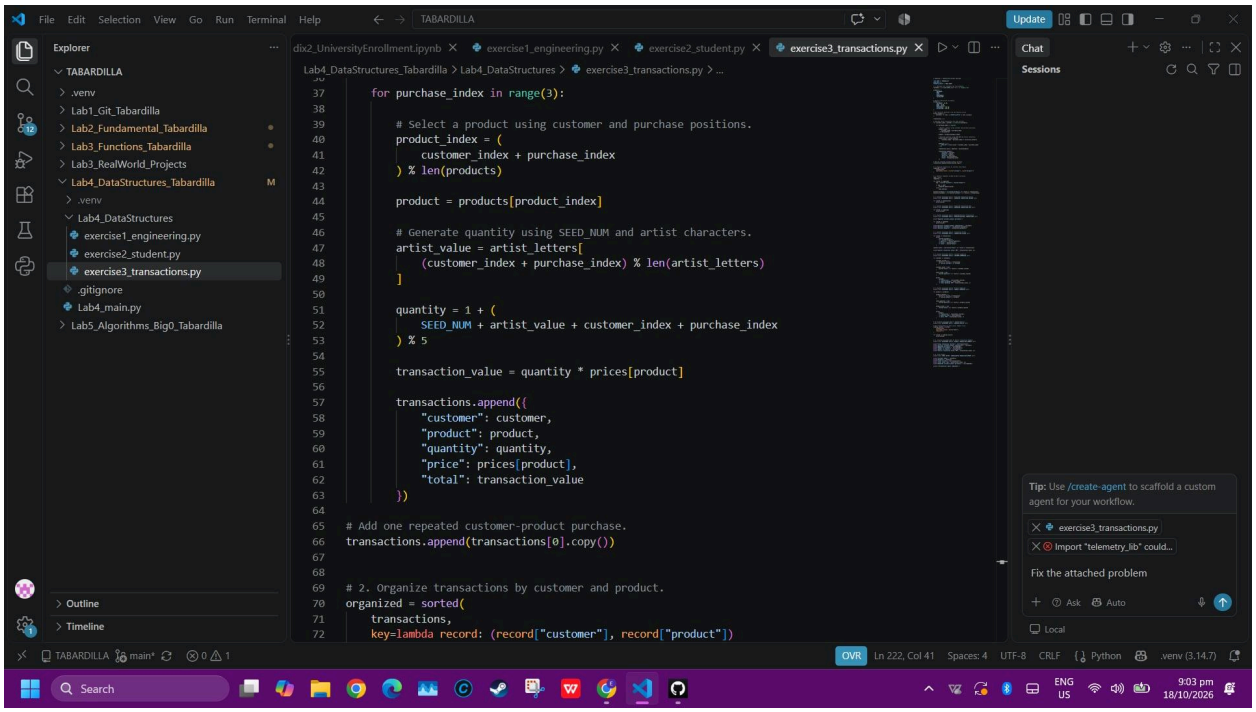

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Problems 1 Output Debug Console Terminal Ports Jupyter
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=== ASSESSMENT DATA 5: STUDENT SUMMARIES ===
TABARDILLA_S1 | Valid scores: 6 | Average: 68.33
TABARDILLA_S2 | Valid scores: 5 | Average: 63.6
TABARDILLA_S3 | Valid scores: 5 | Average: 65.6
TABARDILLA_S4 | Valid scores: 5 | Average: 75.6
TABARDILLA_S5 | Valid scores: 5 | Average: 70.2
Subject summaries:
PROGRAMMING | Valid scores: 6 | Average: 70.17
MATHEMATICS | Valid scores: 5 | Average: 69.8
ELECTRONICS | Valid scores: 5 | Average: 65.6
PHYSICS | Valid scores: 5 | Average: 69.4
ENGLISH | Valid scores: 5 | Average: 68.0
=== ASSESSMENT DATA 6: ORDERED RESULTS ===
{'student': 'TABARDILLA_S4', 'subject': 'MATHEMATICS', 'score': 97, 'category': 'Excellent'}
{'student': 'TABARDILLA_S5', 'subject': 'PROGRAMMING', 'score': 93, 'category': 'Excellent'}
{'student': 'TABARDILLA_S4', 'subject': 'ENGLISH', 'score': 81, 'category': 'Good'}
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{'student': 'TABARDILLA_S4', 'subject': 'PHYSICS', 'score': 73, 'category': 'Needs Improvement'}
{'student': 'TABARDILLA_S1', 'subject': 'PROGRAMMING', 'score': 72, 'category': 'Needs Improvement'}
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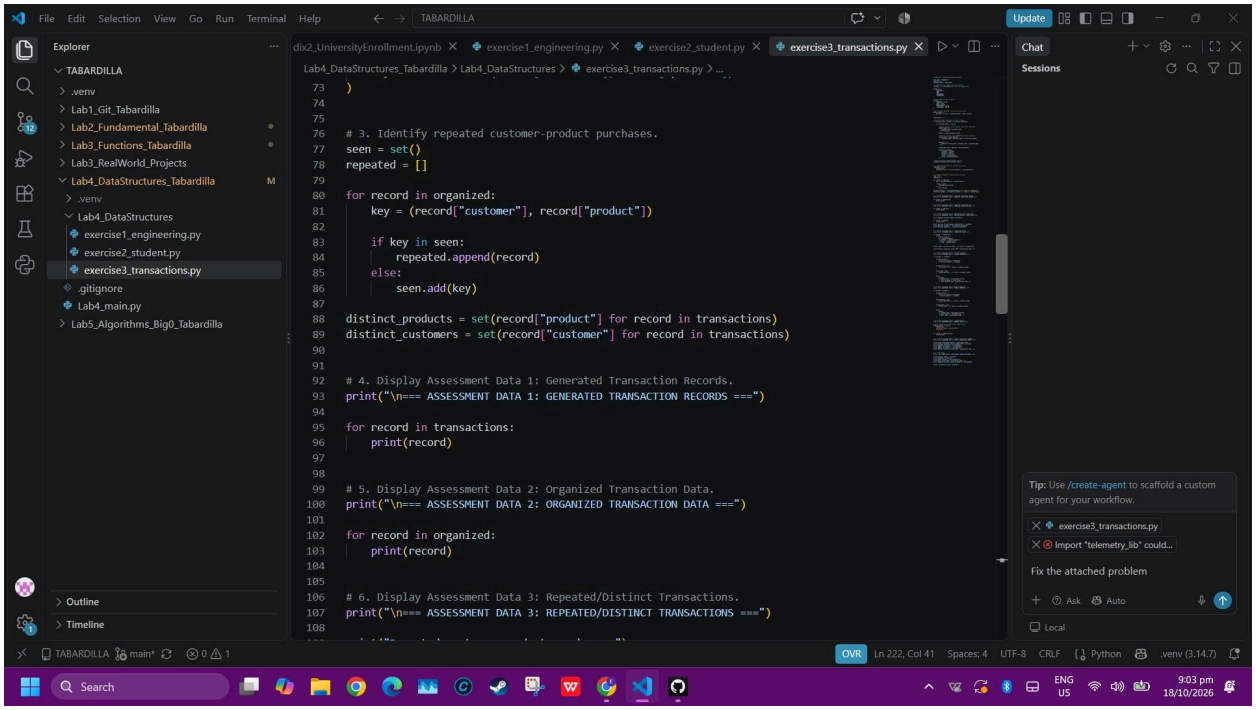


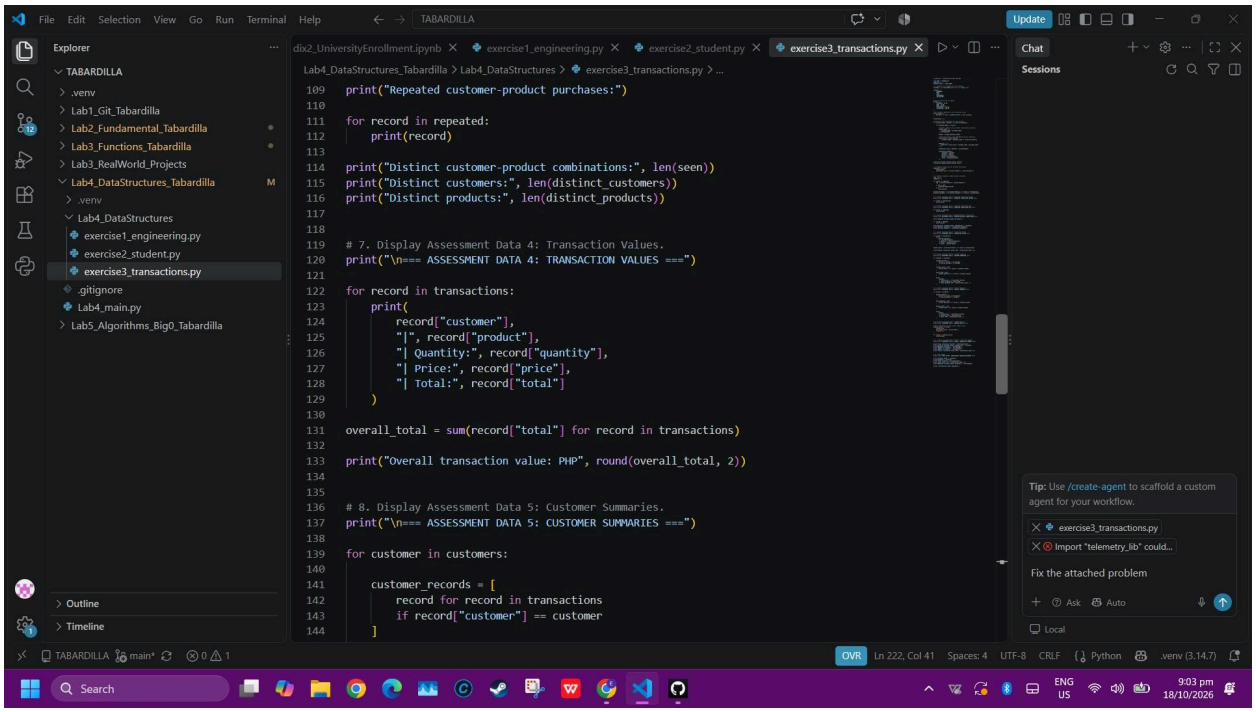


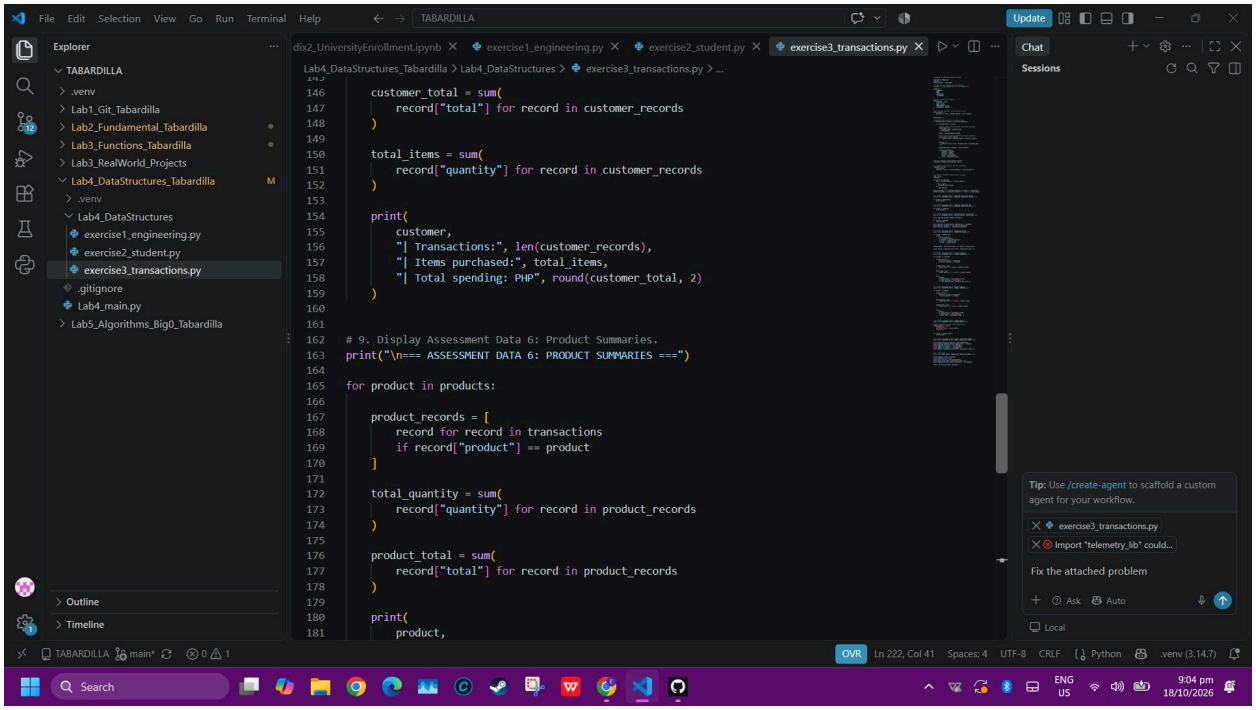
EXERCISE 3: TRANSACTION HISTORY ANALYZER

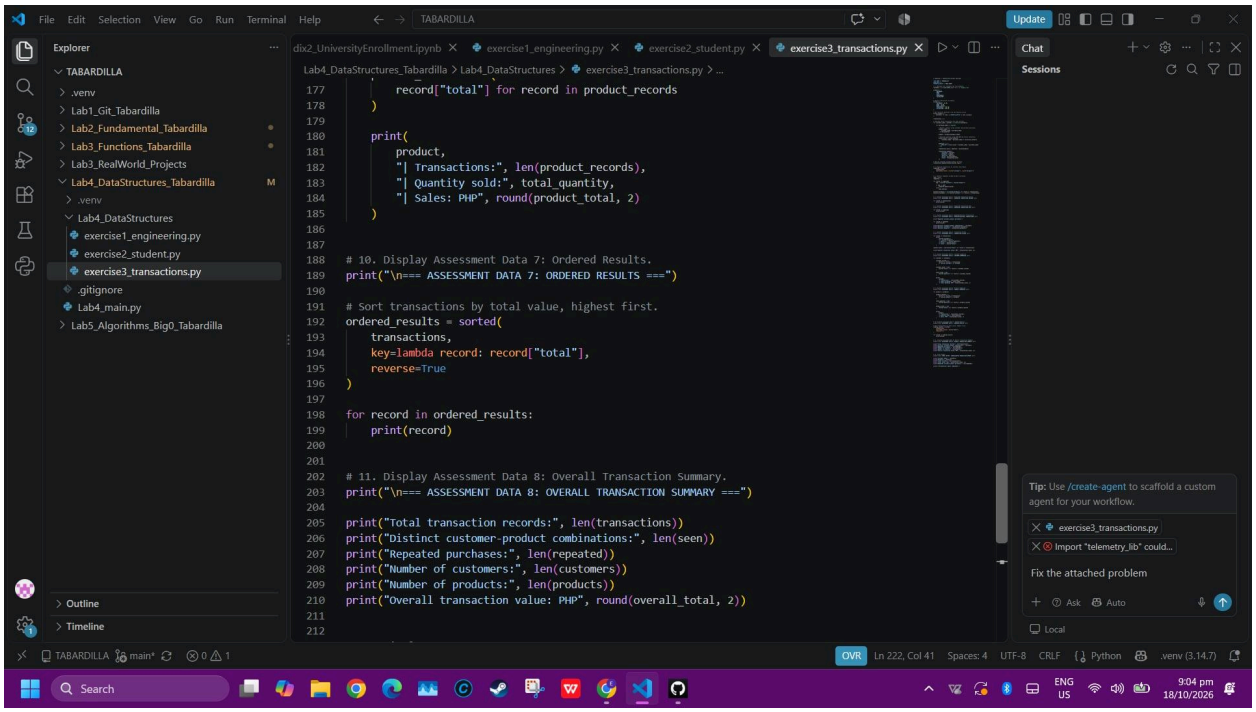

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Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise3_transactions.py > ...
EXERCISE 3: TRANSACTION HISTORY ANALYZER
LAST_NAME = "TABARDILLA"
SEED_NUM = 8
FAVORITE_ARTIST = "NEW JEANS"
1. Generate five customers and five products.
customers = [f"{LAST_NAME}_C{i}" for i in range(1, 6)]
products = [
"NOTEBOOK",
"PEN",
"USB",
"HEADPHONES",
"CALCULATOR"
]
Assign fixed prices (in pesos).
prices = {
"NOTEBOOK": 35.00,
"PEN": 15.00,
"USB": 250.00,
"HEADPHONES": 450.00,
"CALCULATOR": 300.00
}
Get character positions from the favorite artist.
artist_letters = [
ord(char) for char in FAVORITE_ARTIST if char.isalpha()
]
transactions = []
Generate three transactions for each customer.
for customer_index, customer in enumerate(customers):
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise3_transactions.py
Import "telemetry_lib" could...
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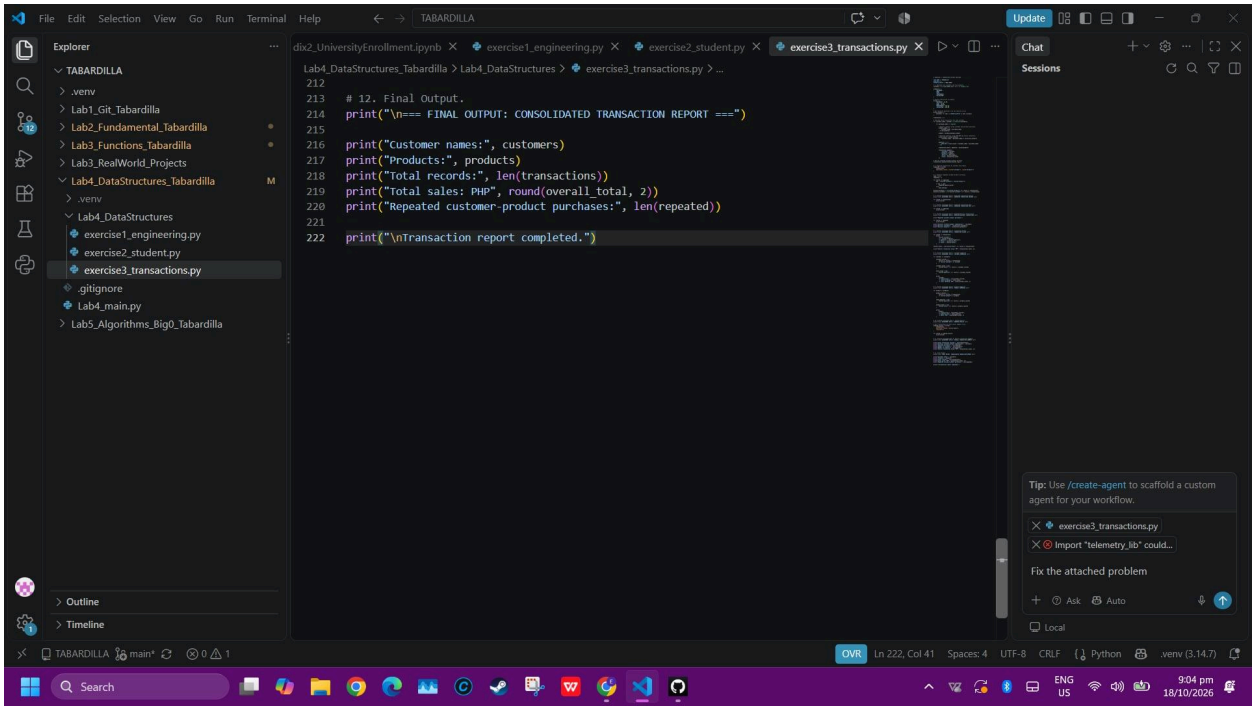

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Lab4_DataStructures_Tabardilla > Lab4_DataStructures > exercise3_transactions.py > ...
for purchase_index in range(3):
Select a product using customer and purchase positions.
product_index = (
customer_index + purchase_index
) % len(products)
product = products[product_index]
Generate quantity using SEED_NUM and artist characters.
artist_value = artist_letters[
(customer_index + purchase_index) % len(artist_letters)
]
quantity = 1 + (
SEED_NUM + artist_value + customer_index + purchase_index
) % 5
transaction_value = quantity * prices[product]
transactions.append({
"customer": customer,
"product": product,
"quantity": quantity,
"price": prices[product],
"total": transaction_value
})
Add one repeated customer-product purchase.
transactions.append(transactions[0].copy())
2. Organize transactions by customer and product.
organized = sorted(
transactions,
key=lambda record: (record["customer"], record["product"])
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Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise3_transactions.py
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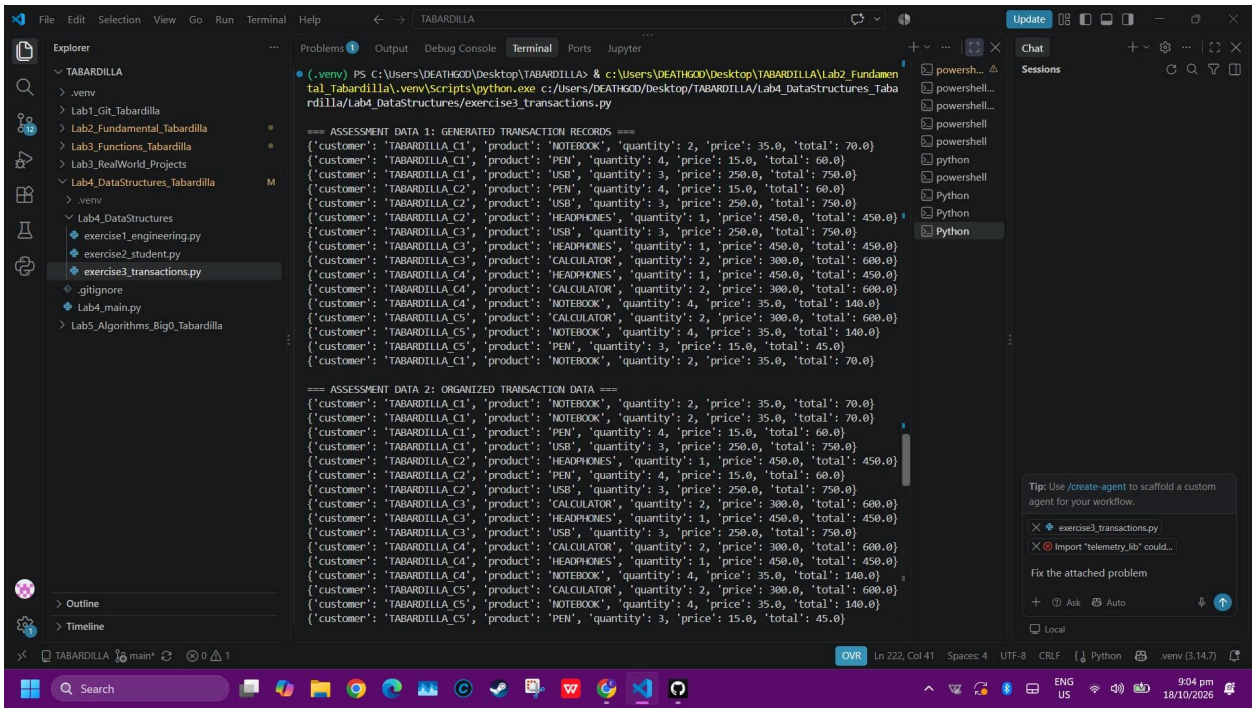

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)
3. Identify repeated customer-product purchases.
seen = set()
repeated = []
for record in organized:
key = (record["customer"], record["product"])
if key in seen:
repeated.append(record)
else:
seen.add(key)
distinct_products = set(record["product"] for record in transactions)
distinct_customers = set(record["customer"] for record in transactions)
4. Display Assessment Data 1: Generated Transaction Records.
print("\n=== ASSESSMENT DATA 1: GENERATED TRANSACTION RECORDS ===")
for record in transactions:
print(record)
5. Display Assessment Data 2: Organized Transaction Data.
print("\n=== ASSESSMENT DATA 2: ORGANIZED TRANSACTION DATA ===")
for record in organized:
print(record)
6. Display Assessment Data 3: Repeated/Distinct Transactions.
print("\n=== ASSESSMENT DATA 3: REPEATED/DISTINCT TRANSACTIONS ===")
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exercise3_transactions.py
Import "telemetry_lib" could...
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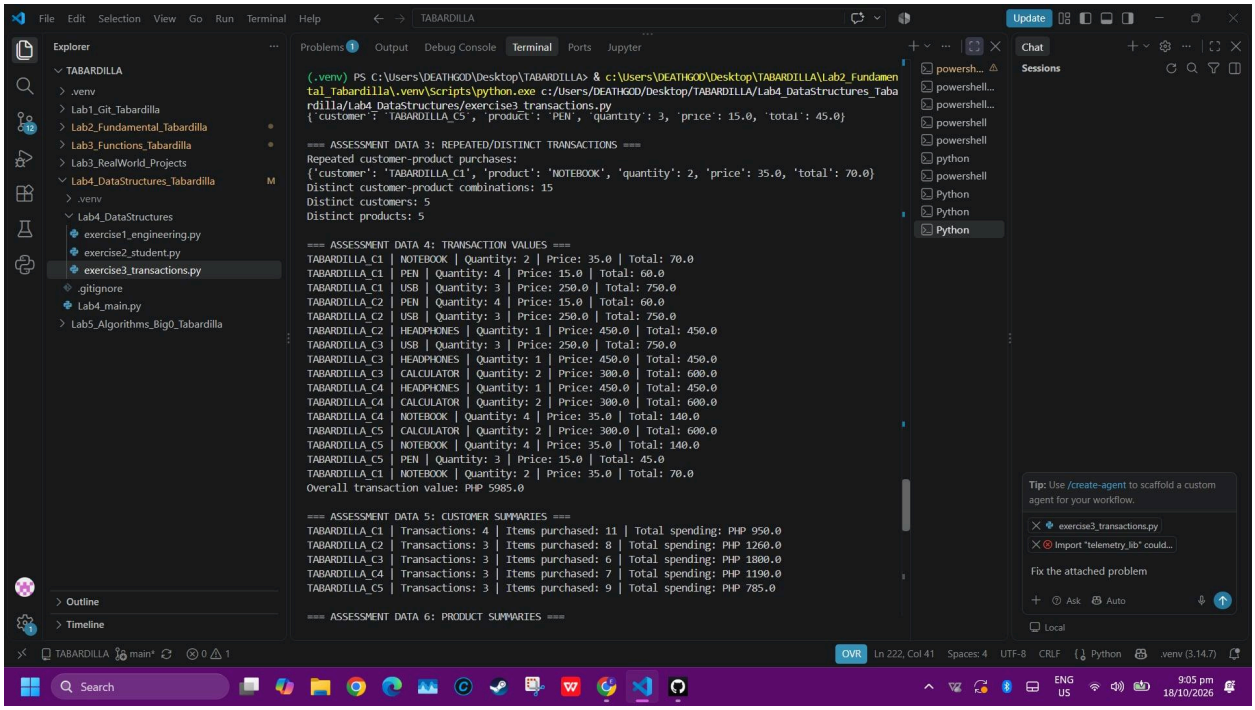

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print("Repeated customer-product purchases:")
for record in repeated:
print(record)
print("Distinct customer-product combinations:", len(seen))
print("Distinct customers:", len(distinct_customers))
print("Distinct products:", len(distinct_products))
7. Display Assessment Data 4: Transaction Values.
print("\n=== ASSESSMENT DATA 4: TRANSACTION VALUES ===")
for record in transactions:
print(
record["customer"],
"|", record["product"],
"| Quantity:", record["quantity"],
"| Price:", record["price"],
"| Total:", record["total"]
)
overall_total = sum(record["total"] for record in transactions)
print("Overall transaction value: PHP", round(overall_total, 2))
8. Display Assessment Data 5: Customer Summaries.
print("\n=== ASSESSMENT DATA 5: CUSTOMER SUMMARIES ===")
for customer in customers:
customer_records = [
record for record in transactions
if record["customer"] == customer
]
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exercise3_transactions.py
Import "telemetry_lib" could...
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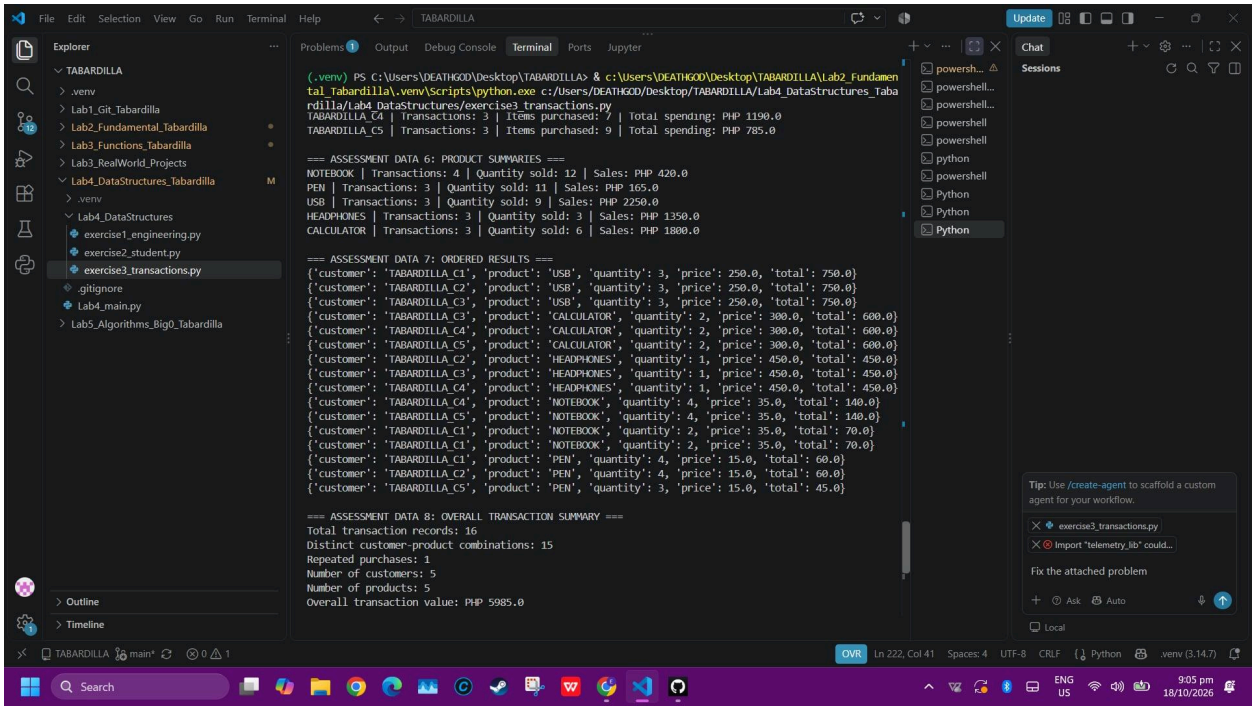

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customer_total = sum(
record["total"] for record in customer_records
)
total_items = sum(
record["quantity"] for record in customer_records
)
print(
customer,
"| Transactions:", len(customer_records),
"| Items purchased:", total_items,
"| Total spending: PHP", round(customer_total, 2)
)
9. Display Assessment Data 6: Product Summaries.
print("\n=== ASSESSMENT DATA 6: PRODUCT SUMMARIES ===")
for product in products:
product_records = [
record for record in transactions
if record["product"] == product
]
total_quantity = sum(
record["quantity"] for record in product_records
)
product_total = sum(
record["total"] for record in product_records
)
print(
product,
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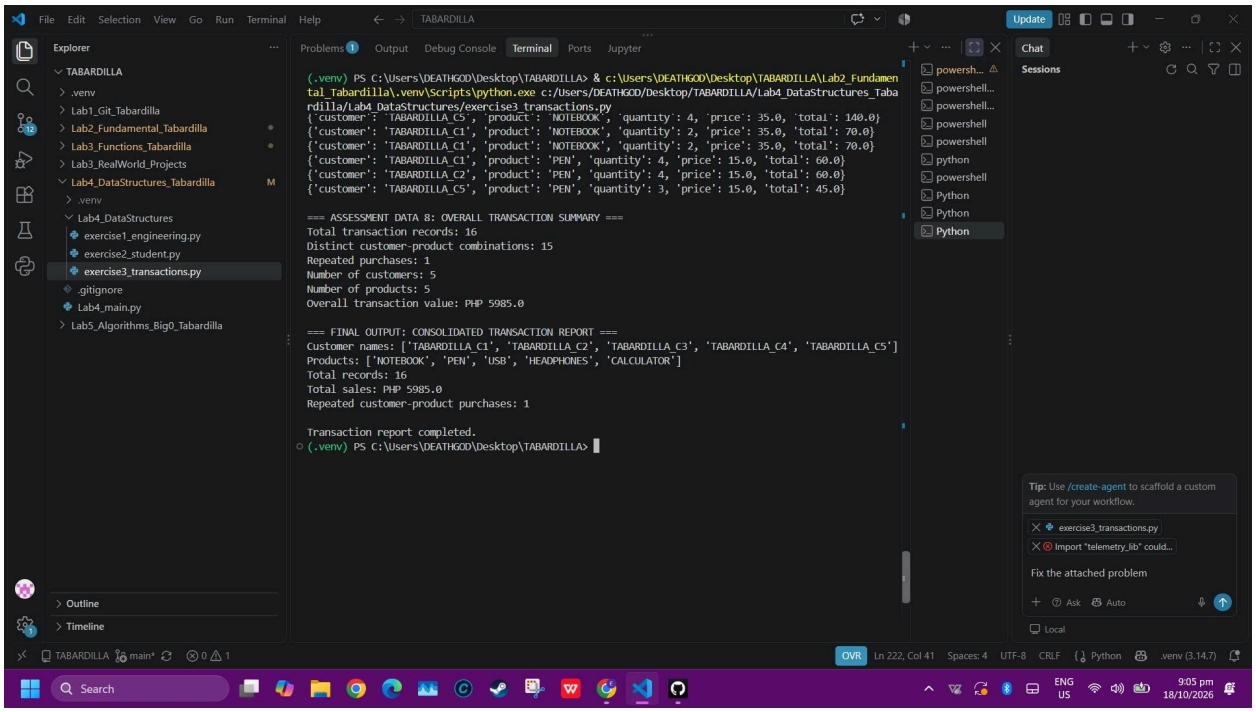

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record["total"] for record in product_records
)
print(
product,
"| Transactions:", len(product_records),
"| Quantity sold:", total_quantity,
"| Sales: PHP", round(product_total, 2)
)
10. Display Assessment Data 7: Ordered Results.
print("\n=== ASSESSMENT DATA 7: ORDERED RESULTS ===")
Sort transactions by total value, highest first.
ordered_results = sorted(
transactions,
key=lambda record: record["total"],
reverse=True
)
for record in ordered_results:
print(record)
11. Display Assessment Data 8: Overall Transaction Summary.
print("\n=== ASSESSMENT DATA 8: OVERALL TRANSACTION SUMMARY ===")
print("Total transaction records:", len(transactions))
print("Distinct customer-product combinations:", len(seen))
print("Repeated purchases:", len(repeated))
print("Number of customers:", len(customers))
print("Number of products:", len(products))
print("Overall transaction value: PHP", round(overall_total, 2))
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212
213 # 12. Final Output.
214 print("\n=== FINAL OUTPUT: CONSOLIDATED TRANSACTION REPORT ===")
215
216 print("Customer names:", customers)
217 print("Products:", products)
218 print("Total records:", len(transactions))
219 print("Total sales: PHP", round(overall_total, 2))
220 print("Repeated customer-product purchases:", len(repeated))
221
222 print("\nTransaction report completed.")
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Tip: Use /create-agent to scaffold a custom agent for your workflow.
exercise3_transactions.py
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Problems 1
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=== ASSESSMENT DATA 1: GENERATED TRANSACTION RECORDS ===
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C2', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C2', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C2', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C3', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C3', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C3', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C4', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C4', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C4', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C5', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C5', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C5', 'product': 'PEN', 'quantity': 3, 'price': 15.0, 'total': 45.0}
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
=== ASSESSMENT DATA 2: ORGANIZED TRANSACTION DATA ===
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C2', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C2', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C2', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C3', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C3', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C3', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C4', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C4', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C4', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C5', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C5', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C5', 'product': 'PEN', 'quantity': 3, 'price': 15.0, 'total': 45.0}
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Tip: Use /create-agent to scaffold a custom agent for your workflow.
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Problems 1 Output Debug Console Terminal Ports Jupyter
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{'customer': 'TABARDILLA_C5', 'product': 'PEN', 'quantity': 3, 'price': 15.0, 'total': 45.0}
=== ASSESSMENT DATA 3: REPEATED/DISTINCT TRANSACTIONS ===
Repeated customer-product purchases:
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
Distinct customer-product combinations: 15
Distinct customers: 5
Distinct products: 5
=== ASSESSMENT DATA 4: TRANSACTION VALUES ===
TABARDILLA_C1 | NOTEBOOK | Quantity: 2 | Price: 35.0 | Total: 70.0
TABARDILLA_C1 | PEN | Quantity: 4 | Price: 15.0 | Total: 60.0
TABARDILLA_C1 | USB | Quantity: 3 | Price: 250.0 | Total: 750.0
TABARDILLA_C2 | PEN | Quantity: 4 | Price: 15.0 | Total: 60.0
TABARDILLA_C2 | USB | Quantity: 3 | Price: 250.0 | Total: 750.0
TABARDILLA_C2 | HEADPHONES | Quantity: 1 | Price: 450.0 | Total: 450.0
TABARDILLA_C3 | USB | Quantity: 3 | Price: 250.0 | Total: 750.0
TABARDILLA_C3 | HEADPHONES | Quantity: 1 | Price: 450.0 | Total: 450.0
TABARDILLA_C3 | CALCULATOR | Quantity: 2 | Price: 300.0 | Total: 600.0
TABARDILLA_C4 | HEADPHONES | Quantity: 1 | Price: 450.0 | Total: 450.0
TABARDILLA_C4 | CALCULATOR | Quantity: 2 | Price: 300.0 | Total: 600.0
TABARDILLA_C4 | NOTEBOOK | Quantity: 4 | Price: 35.0 | Total: 140.0
TABARDILLA_C5 | CALCULATOR | Quantity: 2 | Price: 300.0 | Total: 600.0
TABARDILLA_C5 | NOTEBOOK | Quantity: 4 | Price: 35.0 | Total: 140.0
TABARDILLA_C5 | PEN | Quantity: 3 | Price: 15.0 | Total: 45.0
TABARDILLA_C1 | NOTEBOOK | Quantity: 2 | Price: 35.0 | Total: 70.0
Overall transaction value: PHP 5985.0
=== ASSESSMENT DATA 5: CUSTOMER SUMMARIES ===
TABARDILLA_C1 | Transactions: 4 | Items purchased: 11 | Total spending: PHP 950.0
TABARDILLA_C2 | Transactions: 3 | Items purchased: 8 | Total spending: PHP 1260.0
TABARDILLA_C3 | Transactions: 3 | Items purchased: 6 | Total spending: PHP 1800.0
TABARDILLA_C4 | Transactions: 3 | Items purchased: 7 | Total spending: PHP 1190.0
TABARDILLA_C5 | Transactions: 3 | Items purchased: 9 | Total spending: PHP 785.0
=== ASSESSMENT DATA 6: PRODUCT SUMMARIES ===
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(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA> & c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\python.exe c:/Users/DEATHGOD/Desktop/TABARDILLA/Lab4_DataStructures_Tabardilla/Lab4_DataStructures/exercise3_transactions.py
TABARDILLA_C4 | Transactions: 3 | Items purchased: 7 | Total spending: PHP 1190.0
TABARDILLA_C5 | Transactions: 3 | Items purchased: 9 | Total spending: PHP 785.0
=== ASSESSMENT DATA 6: PRODUCT SUMMARIES ===
NOTEBOOK | Transactions: 4 | Quantity sold: 12 | Sales: PHP 420.0
PEN | Transactions: 3 | Quantity sold: 11 | Sales: PHP 165.0
USB | Transactions: 3 | Quantity sold: 9 | Sales: PHP 2250.0
HEADPHONES | Transactions: 3 | Quantity sold: 3 | Sales: PHP 1350.0
CALCULATOR | Transactions: 3 | Quantity sold: 6 | Sales: PHP 1800.0
=== ASSESSMENT DATA 7: ORDERED RESULTS ===
{'customer': 'TABARDILLA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C2', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C3', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'TABARDILLA_C3', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C4', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C5', 'product': 'CALCULATOR', 'quantity': 2, 'price': 300.0, 'total': 600.0}
{'customer': 'TABARDILLA_C2', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C3', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C4', 'product': 'HEADPHONES', 'quantity': 1, 'price': 450.0, 'total': 450.0}
{'customer': 'TABARDILLA_C4', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C5', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C2', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C5', 'product': 'PEN', 'quantity': 3, 'price': 15.0, 'total': 45.0}
=== ASSESSMENT DATA 8: OVERALL TRANSACTION SUMMARY ===
Total transaction records: 16
Distinct customer-product combinations: 15
Repeated purchases: 1
Number of customers: 5
Number of products: 5
Overall transaction value: PHP 5985.0
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Fix the attached problem


(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA> & c:\Users\DEATHGOD\Desktop\TABARDILLA\Lab2_Fundamental_Tabardilla\.venv\Scripts\python.exe c:/Users/DEATHGOD/Desktop/TABARDILLA/Lab4_DataStructures_Tabardilla/Lab4_DataStructures/exercise3_transactions.py
{'customer': 'TABARDILLA_C5', 'product': 'NOTEBOOK', 'quantity': 4, 'price': 35.0, 'total': 140.0}
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'TABARDILLA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C2', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'TABARDILLA_C5', 'product': 'PEN', 'quantity': 3, 'price': 15.0, 'total': 45.0}
=== ASSESSMENT DATA 8: OVERALL TRANSACTION SUMMARY ===
Total transaction records: 16
Distinct customer-product combinations: 15
Repeated purchases: 1
Number of customers: 5
Number of products: 5
Overall transaction value: PHP 5985.0
=== FINAL OUTPUT: CONSOLIDATED TRANSACTION REPORT ===
Customer names: ['TABARDILLA_C1', 'TABARDILLA_C2', 'TABARDILLA_C3', 'TABARDILLA_C4', 'TABARDILLA_C5']
Products: ['NOTEBOOK', 'PEN', 'USB', 'HEADPHONES', 'CALCULATOR']
Total records: 16
Total sales: PHP 5985.0
Repeated customer-product purchases: 1
Transaction report completed.
(.venv) PS C:\Users\DEATHGOD\Desktop\TABARDILLA>
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Fix the attached problem